

Hull Slamming

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*This report presents an in-depth review of the current state of knowledge on hull slamming, which is one of several types of slamming problems to be considered in the design and operation of ships. Hull slamming refers to the impact of the hull or a section of the hull as it reenters the water. It can be considered to be part of a larger class of water entry problems that include the water landing of spacecraft and solid rocket boosters, the water landing and ditching of aircraft, ballistic impacts on fuel tanks, and other applications. The problem involves the interaction of a structure with a fluid that has a free surface. Significant simplifications can be achieved by considering a two-dimensional cross section of simple shape (wedge, cone, sphere, and cylinder) and by assuming that the structure is a rigid body. The water is generally modeled as an incompressible, irrotational, inviscid fluid. Two approximate solutions developed by von Karman (1929, "The Impact on Seaplane Floats During Landing," NACA Technical Note NACA-TN-32) and Wagner (1932, "Über stoss und Gleitvorgänge an der Oberache von Flüssigkeiten," Z. Angew. Math. Mech., **12**, pp. 192–215) can be used to predict the motion of the body, the hydrodynamic force, and the pressure distribution on the wetted surface of the body. Near the intersection with the initial water surface, water piles up, a jet is formed, and the solution has a singularity in this region. It was shown that nearly half of the kinetic energy transferred from the solid to the fluid is contained in this jet, the rest being stored in the bulk of the fluid. A number of complicating factors are considered, including oblique or asymmetric impacts, elastic deformations, and more complex geometries. Other marine applications are considered as well as applications in aerospace engineering. Emphasis is placed on basic principles and analytical solutions as an introduction to this topic, but numerical approaches are needed to address practical problems, so extensive references to numerical approaches are also given.*

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1 Introduction

Composite materials have long been used in marine structures. In their comprehensive review of the use of composite structures for naval ships and submarines, Mouritz et al. [1] indicated that the use of composite materials started immediately after World War II. Initially, this use was restricted to small crafts and parts of larger vessels. The construction of large all-composite naval vessels started in the 1970s with, for example, the HMS Wilton, a world leader at the time of her construction in 1973. At 450 tons, she was then the world's largest reinforced plastic ship [2]. Submarines also make significant use of composite materials. For example, the Virginia class submarines have a 25-ton composite bow dome that houses and protects the sonar transducer sphere. The bow dome is 21 feet long and has a 26-foot diameter at the attachment end.

More recently, large vessels designed to reach speeds of up to 50 knots have been built using composite sandwich structures. Two examples are frequently mentioned: (1) the Norwegian patrol boat Skjold and (2) the Swedish corvette Visby. The 72-m-long Visby, with a displacement of 650 mt and a top speed of 38 knots, is the largest commercial or combatant ship hull entirely constructed of carbon fiber-reinforced plastics (CFRP) using a sandwich construction with a polyvinyl chloride (PVC) foam core and vinyl ester resin.

Ships are subjected to four types of water impacts that are commonly called slamming. The first is called hull slamming and occurs when the hull reenters the water after having been partially or completely lifted out of the water. The other three types of slamming events involve the impact of water waves on ships or other marine structures: (1) bow slamming occurs when a wave hits the bow of a ship; (2) green water slamming occurs when a wave hits the deck or a wall of the upper deck structure of a ship; and (3) wet-deck slamming occurs when a wave hits the underside

of the deck joining two hulls of a multihull vessel. This article will focus on hull slamming. While slamming refers to problems in which water impacts the outside of a structure, the term sloshing designates a type of related problem, in which fluid inside a container is undergoing violent motions and generating significant loads on the walls of that container. Sloshing problems are discussed in detail in a recent book [3] and are not considered here.

The impact of a rigid body entering a body of water has been studied for many years. In the late 1800s, Worthington used time lapse photography to provide insights into the details of the impact of a drop of water into milk [4] and the impact of a sphere on water [5]. Von Karman [6] in 1929 and Wagner [7] in 1932 presented the first analytical studies of the water entry problem in connection with the practical problem of water landing of sea planes. During the emergency operation called airplane ditching, like the ditching of an Airbus on the Hudson river in New York City on January 15, 2009, the plane approaches the surface of the water with a low vertical velocity and the tail of the plane penetrates the water and generates a vertical force, a drag force, and a bending moment on the fuselage. During the Apollo space missions, capsules carrying the astronauts ended their flight by a water landing after parachutes had deployed to reduce the entry velocity. Analyzing the dynamics of this water entry is important to ensure the safety of the occupants. The solid rocket boosters used during the launch of the space shuttle are released after the fuel is spent. Soon after, parachutes are deployed and they also land on water. Those are four examples of water entry problems in aerospace engineering, and a detailed review of the literature in this area was published recently [8]. This review will also address the problem of a falling cylinder in water as it relates to a mine dropped from a helicopter or a ship and drops to the bottom of the ocean, where it is expected to bury itself into the sediment. Ballistic impacts into fluid-filled tanks and the motion of underwater supercavitating projectiles will also be addressed.

In general terms, the water entry of a projectile consists of a cavity-forming phase, an open cavity phase, and a closed cavity phase [9] (Fig. 1). Initially, the cavity is filled by air, and after a

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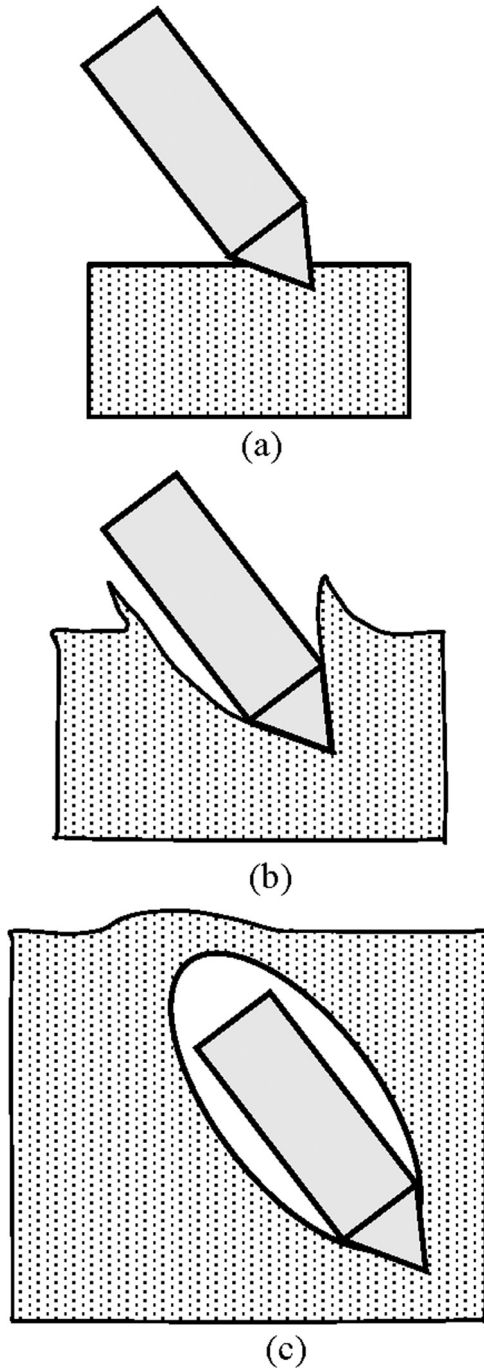


Fig. 1 Three phases in the impact of a projectile in water (Mackey [9]): (a) flow formation, (b) open cavity (or cavity running), (c) closed cavity

certain penetration depth, the cavity closes and is still filled with air. Later, that cavity detaches from the projectile, which will then be completely surrounded by water or, if the velocities are large enough and pressure drop below a certain level, a new cavity will form, containing water vapor. This is called cavitation and, if the projectile is completely enclosed by the cavity, it is called a supercavitating projectile. Hull slamming is usually restricted to the cavity-forming phase; some of the other problems considered here involve all possible scenarios.

2 Water Entry of a Rigid Body

In this section, we first discuss the various forces acting on a rigid body impacting the free surface of a fluid and their relative

importance. Methods developed by von Karman [6] and Wagner [7] are discussed next, since they are widely used and provide insight into the dynamics of the impact.

2.1 Equations of Motion. The external forces acting on a rigid body entering the water are the weight Mg , the buoyancy force F_B , the steady state drag force F_D , the capillary force F_c due to surface tension, and the impact force F , also called the hydrodynamic force. Using Newton's second law, the equation of motion of the penetrating body can be written as

$$M\ddot{\zeta} = Mg - F - F_B - F_c - F_D \quad (2.1)$$

where ζ is the penetration depth relative to the undisturbed water surface. The force applied by the fluid on the wetted surface of the body is equal and opposite to the force applied by the body on the fluid. It is common to consider a mass m of fluid that moves with the body, which is usually called the added mass or virtual mass of the fluid. This mass increases during the impact, and the motion of that mass is governed by

$$F = \frac{d}{dt}(m\dot{\zeta}) = m\ddot{\zeta} + \frac{dm}{d\zeta}\dot{\zeta}^2 \quad (2.2)$$

Substituting into Eq. (2.1) gives

$$\frac{d}{dt}[(M+m)\dot{\zeta}] = Mg - F_B - F_c - F_D \quad (2.3)$$

or

$$(M+m)\ddot{\zeta} = Mg - F_B - F_c - F_D - \frac{dm}{d\zeta}\dot{\zeta}^2 \quad (2.4)$$

The steady state drag force can be written in terms of the fluid density ρ , the cross-sectional area A at the intersection with the water surface, and the steady state drag coefficient C_{ds} as

$$F_D = \frac{1}{2}\rho AC_{ds}\dot{\zeta}^2 \quad (2.5)$$

The instantaneous state drag coefficient C_{di} is defined so that $(dm/d\zeta)\dot{\zeta}^2 = (1/2)\rho AC_{di}\dot{\zeta}^2$. That is,

$$C_{di} = \frac{2}{\rho A} \frac{dm}{d\zeta} \quad (2.6)$$

The weight of the body is written in terms of, $\bar{\rho}_B$, the effective density of the body and its volume, V , as $Mg = V\bar{\rho}_B g$. The buoyancy force is $F_B = V'\rho g$, where V' is the volume of the immersed portion of the body. The capillary force is proportional to, γ , the surface tension and, P , the length of the perimeter of the intersection with the free surface of the water. It depends also on the angle, ϕ , that the tangent to the surface of the water makes with the horizontal. The capillary force can be written as $F_c = \gamma P \sin \phi$. Then, the motion is governed by

$$\frac{d}{dt}[(M+m)\dot{\zeta}] = Mg - F_B - \gamma P \sin \phi - \frac{1}{2}\rho AC_{ds}\dot{\zeta}^2 \quad (2.7)$$

$$(M+m)\ddot{\zeta} = Mg - F_B - \gamma P \sin \phi - \frac{1}{2}\rho A(C_{ds} + C_{di})\dot{\zeta}^2 \quad (2.8)$$

Equation (2.8) is used by Aristoff and Bush [10], for example.

2.2 Relative Importance of the Various Forces. Several of the forces acting on the body and included in Eq. (2.1) can often be neglected for hull slamming problems, while they may be

significant in other applications. This subsection will show that, in general, the effect of viscosity and gravity on the dynamics of the fluid can be neglected as well as the effect of surface tension. In the process, three nondimensional numbers are introduced (Reynolds' number, Weber's number, and Froude's number) to compare the inertia forces to the viscous, capillary, and gravitational forces.

2.2.1 Effect of Fluid Viscosity and Gravity. For a Newtonian fluid, the stress tensor can be written as

$$\sigma_{ij} = -p\delta_{ij} + \mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \quad (2.9)$$

where p is the pressure, $\bar{v}$ is the velocity vector, and μ is the viscosity of the fluid. For inviscid fluids, $\mu = 0$, so that the stress on any surface is normal to the surface and equal to $-p\bar{n}$. The Navier–Stokes equations for an incompressible fluid are

$$\rho \left[\frac{\partial \bar{v}}{\partial t} + (\bar{v} \cdot \nabla) \bar{v} \right] = -\nabla p + \mu \nabla^2 \bar{v} - \rho g \bar{k} \quad (2.10)$$

where $\nabla = \bar{i}(\partial/\partial x) + \bar{j}(\partial/\partial y) + \bar{k}(\partial/\partial z)$, $\nabla p = \nabla \cdot (p\bar{I}) = \bar{i}(\partial p/\partial x) + \bar{j}(\partial p/\partial y) + \bar{k}(\partial p/\partial z)$, $\bar{I}$ is the identity tensor, and $\nabla^2 = \nabla \cdot \nabla$. The right-hand side of Eq. (2.10) accounts for the effect of pressure, viscosity, and gravity of the fluid. In order to assess the relative importance of those three factors, we define the nondimensional velocity, pressure, time, and gradient operator as

$$\bar{v}' = \frac{\bar{v}}{V_o}, \quad p' = \frac{p}{\rho V_o^2}, \quad t' = V_o t/D, \quad \text{and} \quad \nabla' = D \nabla \quad (2.11)$$

where V_o is the initial velocity of the body and D is a characteristic dimension of the body. The nondimensional form of the equations of motion

$$\frac{\partial \bar{v}'}{\partial t'} + (\bar{v}' \cdot \nabla') \bar{v}' = -\nabla' p' + \frac{1}{Re} \nabla'^2 \bar{v}' - \frac{1}{Fr^2} \bar{k} \quad (2.12)$$

which introduces two nondimensional parameters: Reynolds number $Re = (\rho V_o D)/\mu$ and Froude's number $Fr = V_o/\sqrt{gD}$. For large values of Re and Fr , the viscous and gravity effects become negligible. For water, the density $\rho = 1000 \text{ kg/m}^3$, and the viscosity $\mu = 10^{-6} \text{ m}^2/\text{s}$. Taking typical values $D = 1 \text{ m}$ and $V_o = 10 \text{ m/s}$, we find that $Re = 1 \times 10^{10}$ and $Fr = 10.2$. This simple analysis indicates that it is appropriate to neglect viscous and gravity effects in the fluid—a frequently made assumption.

The impact of spheres or cylinders on other types of fluids has been considered by several authors. For example, Akers and Belmonte [11] studied the impact of a sphere on a viscoelastic micellar fluid. Tabuteau et al. [12] examined the impact on a gel that behaves as a fluid with yield stress. These types of fluids are not considered in this work.

2.3 Surface Tension. Many animals, such as insects, spiders, birds, lizards, and mammals, walk on water. These water walkers range in size from less than 1 mm to 3 m for the tail-walking dolphin [13]. Several articles describe the water strider locomotion of insects [14–20] and of basilisk lizards [21–24]. In many of these cases, surface tension plays a significant role in enabling the animal to stand and/or move on the surface of the water. Consider a segment of unit length of a circular cylinder of radius r floating in a liquid (Fig. 2). The force per unit length f is balanced by the vertical components of the two surface tension forces and the buoyancy force.

During the water entry of a solid, the resistance to the motion due to fluid inertia effects is proportional to ρV^2 and the projected area, which varies with D^2 . The force due to surface tension effects depends on σ and the length of the circumference, so it is

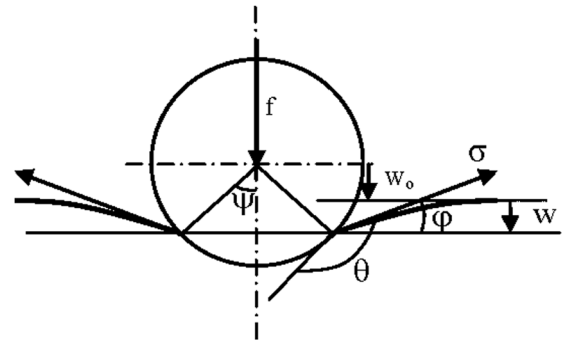


Fig. 2 Cylinder of unit length floating on a liquid. The external force per unit length f is balanced by the vertical components of the two surface tension forces σ . The geometry is defined by the wetting angle θ .

proportional to σD . The relative importance of these effects is estimated by Weber number defined as the ratio

$$We = \frac{\rho V_o^2 D^2}{\sigma D} = \frac{\rho V_o^2 D}{\sigma} \quad (2.13)$$

where σ is the surface tension. The Weber number is useful in the dynamic case. In the static case, the force due to surface tension is compared to the weight of the object, which is proportional to $\rho g D^3$ using the Bond number defined as ratio

$$Bo = \frac{\rho g D^3}{\sigma D} = \frac{\rho g D^2}{\sigma} \quad (2.14)$$

If $\sigma = 0.07 \text{ kg/s}^2$, $V = 10 \text{ m/s}$, and $D = 1 \text{ m}$, $We = 1.429 \times 10^6$ and $Bo = 1.401 \times 10^5$. Therefore, surface tension is insignificant when considering hull slamming problems for ships and also when considering the equilibrium of the ship in the static case.

Only small insects can be supported by surface tension. For a man with a mass of 70 kg to be supported by surface tension, he would need to have feet with a perimeter $P = mg/\sigma$ of about 9541 m, as pointed out by Bush and Hu [13]. Some animals must have other means for generating vertical forces to balance the weight. One suggestion is that a vertical force can be generated by slapping feet on the surface of the water. Hsieh [21] and Hsieh and Lauder [22] showed that, for basilisk lizards running across the water surface, the stride consists of slapping (14% of the total duration), the stroke (18%), and the recovery (68%).

2.4 Von Karman's Approach. Assuming that all other forces are negligible, Eq. (2.3) indicates that the motion is governed by

$$\frac{d}{dt} [(M + m)\dot{\zeta}] = 0 \quad (2.15)$$

Integrating with respect to time, keeping in mind that, for $t = 0$, the added mass of fluid is zero ($m = 0$) and $\dot{\zeta}(0) = V_o$, the impact velocity

$$MV_o = (M + m)\dot{\zeta} \quad \text{or} \quad \dot{\zeta} = \frac{MV_o}{M + m} \quad (2.16)$$

Equation (2.16) expresses the conservation of linear momentum throughout the water entry process and is cited by many authors. The problem is illustrated in Fig. 3 for the water entry of a wedge, where the cross-hatched area shows the added mass of the fluid. The velocity of the impacting body decreases as it penetrates the free surface of the fluid and the added mass m increases. Energy

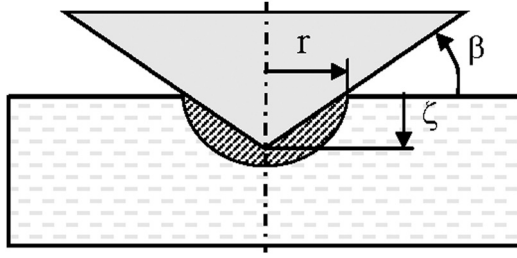


Fig. 3 Von Karman's momentum approach. Penetration depth ζ and deadrise angle β .

is transferred from the solid to the fluid. The change in kinetic energy of the projectile is

$$\Delta T_p = \frac{1}{2} M (v_o^2 - \dot{\zeta}^2) = \frac{1}{2} M v_o^2 \frac{m(2M + m)}{(M + m)^2} \quad (2.17)$$

The change in kinetic energy of the water

$$\Delta T_w = \frac{1}{2} m \dot{\zeta}^2 = \frac{1}{2} M v_o^2 \frac{Mm}{(M + m)^2} \quad (2.18)$$

The ratio of these two quantities

$$\frac{\Delta T_w}{\Delta T_p} = \frac{M}{2M + m} \quad (2.19)$$

shows that less than 50% of the kinetic energy lost by the projectile is transferred to the fluid. References [25–34] have shown that the energy lost by the projectile is absorbed by the bulk of the fluid and by the fluid in the high speed jets that develops at the intersection with the free water surface. These analyses show that, for short times, this energy is split equally between the bulk of the fluid and the jets. This result is also asserted by Payne [35].

2.5 Wagner's Approach. The hydrodynamic force is obtained by integrating the effect of the pressure acting on S_w , the wetted surface of the body

$$\bar{F} = - \int_{S_w} p \bar{n} dS \quad (2.20)$$

where $\bar{n}$ is the outside normal to the body for a small surface element dS . Fluid flow is governed by the conservation of momentum equation

$$\rho \left(\frac{\partial \bar{v}}{\partial t} + (\bar{v} \cdot \nabla) \bar{v} \right) = -\nabla p + \mu \nabla^2 \bar{v} + \bar{F} \quad (2.21)$$

and the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{u}) = 0 \quad (2.22)$$

where $\bar{v}$ is the velocity vector, ρ is the density, p is the pressure, μ is the viscosity of the fluid, and $\bar{F}$ represents the body force that may be applied. If the y -axis is taken in to be in the vertical direction, the effect of gravity is represented by $\bar{f} = \rho g \bar{j}$. For irrotational flows, $(\nabla \times \bar{v} = 0)$ and the velocity can be written in terms of a potential function ϕ

$$\bar{v} = \nabla \phi = \frac{\partial \phi}{\partial x} \bar{i} + \frac{\partial \phi}{\partial y} \bar{j} + \frac{\partial \phi}{\partial z} \bar{k} \quad (2.23)$$

If the fluid is incompressible, the continuity equation becomes $\nabla \cdot \bar{v} = 0$, and for irrotational flow, using Eq. (2.23),

$$\nabla \cdot \bar{v} = \nabla \cdot (\nabla \phi) = \nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (2.24)$$

which is known as Laplace's equation. Substituting Eq. (2.23) into Eq. (2.21) and neglecting the effects of viscosity and gravity leads to Bernoulli's equation,

$$\frac{p}{\rho} = -\frac{\partial \phi}{\partial t} - \frac{1}{2} (\nabla \phi \cdot \nabla \phi) \quad (2.25)$$

relates the pressure p to the velocity potential ϕ . Using Bernoulli's equation, the force $\bar{F}$ acting on the body (Eq. (2.20)) becomes

$$\bar{F} = -\rho \int_{S_w} \left(\frac{\partial \phi}{\partial t} + \frac{1}{2} (\nabla \phi \cdot \nabla \phi) \right) \bar{n} dS \quad (2.26)$$

The problem is then to find a potential function that satisfies Laplace's equation and the boundary conditions. The boundary of the fluid domain consists of S_w , the wetted surface of the impacting body, and the free surface, S_F . The shape and position of the free surface during water entry is not known, and consequently, neither is the boundary of the wetted surface of the body. On S_w , the velocity of the solid and that of the fluid are the same in the normal direction

$$\bar{v} \cdot \bar{n} = \bar{V} \cdot \bar{n} \quad (2.27)$$

where $\bar{V}$ is the velocity of the solid at that point. The velocity of the fluid in the direction normal to the wetted surface can be written as $\bar{v} \cdot \bar{n} = (\partial \phi / \partial n)$. On S_F , the pressure must be zero, so

$$\frac{\partial \phi}{\partial t} + \frac{1}{2} (\nabla \phi \cdot \nabla \phi) = 0 \quad (2.28)$$

This general discussion shows that finding a potential function that satisfies Laplace's equation and the boundary conditions is a challenge, particularly since the boundaries are not known.

In Wagner's two-dimensional approach, the body is replaced by a flat plate of length $2r$, which is the intersection of the body with the initial plane of the water. As the penetration depth increases, r also increases. This is why this model is called Wagner's expanding plate model. In two dimensions, the velocity vector is given by

$$\bar{v} = \nabla \phi = \frac{\partial \phi}{\partial x} \bar{i} + \frac{\partial \phi}{\partial y} \bar{j} \quad (2.29)$$

and in Wagner's model, the potential is taken as

$$\phi = -V \sqrt{r^2 - z^2} \quad (2.30)$$

where $z = x + iy$. Then, the horizontal component of the velocity along the plate ($|x| < r, y = 0$) is

$$\frac{\partial \phi}{\partial x}(x, 0) = V \frac{x}{\sqrt{r^2 - x^2}} \quad (2.31)$$

and the vertical component of the velocity is zero along the bottom surface of the plate. Note that the velocity of the fluid is zero in the center ($x = y = 0$) and that it becomes infinite near the edge ($x = r, y = 0$). The pressure is obtained using the Bernoulli equation (Eq. (2.25)). Therefore, along the bottom of the plate,

$$\frac{p}{\rho} = \frac{\partial}{\partial t} \left(V \sqrt{r^2 - x^2} \right) - \frac{1}{2} \left(V \frac{x}{\sqrt{r^2 - x^2}} \right)^2 \quad (2.32)$$

Recalling that both the velocity of the impacting body and r , the length of the plate in the Wagner model, vary with time,

$$\frac{p}{\rho} = \dot{V}\sqrt{r^2 - x^2} + \frac{V\dot{r}}{\sqrt{r^2 - x^2}} - \frac{1}{2} \frac{V^2 x^2}{r^2 - x^2} \quad (2.33)$$

Examining the three terms on the right-hand side of Eq. (2.33), we see that, as $x \rightarrow r$, the first term tends to zero and the other two terms become infinite. Taking the ratio of the second and third terms,

$$\left(\frac{V\dot{r}}{\sqrt{r^2 - x^2}} \right) / \left(\frac{1}{2} \frac{V^2 x^2}{r^2 - x^2} \right) = \frac{2\dot{r}}{Vx^2} \sqrt{r^2 - x^2} \quad (2.34)$$

we see that, as $x \rightarrow r$, the third term becomes much larger than the second, so that the pressure becomes negative near the edge of the wetted surface.

3 Water Entry of a Wedge

Simple geometric shapes are often used to obtain analytical solutions or to conduct experiments and gain some insight into the mechanics of the water impact of a rigid body. Wedges were used by von Karman [6] and Wagner [7], because they resembled the shape of floats on seaplanes. Wedges also approximate the shape of a ship's cross-section. In this section, we use von Karman's approach to study the dynamics of the impact and calculate the hydrodynamic force acting on the wedge. Then, Wagner's method is used to determine the pressure distribution along the wetted edge, and we discuss the singularity near the intersection with the free surface. Finally, we discuss several complicating factors: spray and jet formation, air cushioning, oblique and asymmetric impacts, water compressibility, and elastic wedge impacts.

3.1 Von Karman's Approach for a Wedge. Von Karman considered a wedge of unit thickness, mass M , and deadrise angle β entering the water with a vertical velocity V_o (Fig. 3). As the wedge penetrates the initial free surface of the water, it is assumed that the mass of a half disk of water of unit width with radius r is moving with the wedge. This added mass m is given by

$$m = \frac{\pi}{2} \rho r^2 = \frac{\pi}{2} \rho \frac{\zeta^2}{\tan^2 \beta} \quad (3.1)$$

Since the water piles up near the intersection point, the added mass should be calculated as shown in Fig. 4 which implies that ζ should be replaced by $\zeta\gamma$ in Eq. (3.1). Then,

$$m = \frac{\pi}{2} \rho r^2 = \frac{\pi}{2} \rho \frac{\gamma^2 \zeta^2}{\tan^2 \beta} \quad (3.2)$$

The value of the coefficient γ will be discussed later. Using the conservation of momentum equation (Eq. (2.14)), the velocity of the wedge

$$\dot{\zeta} = \frac{V_o}{1 + \frac{\pi}{2} \rho \frac{\gamma^2 \zeta^2}{M \tan^2 \beta}} \quad (3.3)$$

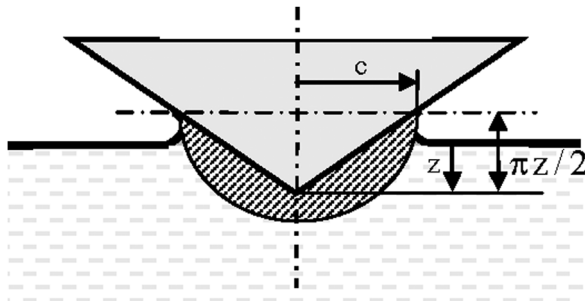


Fig. 4 Water pile-up near the edge, as estimated by Wagner [7]

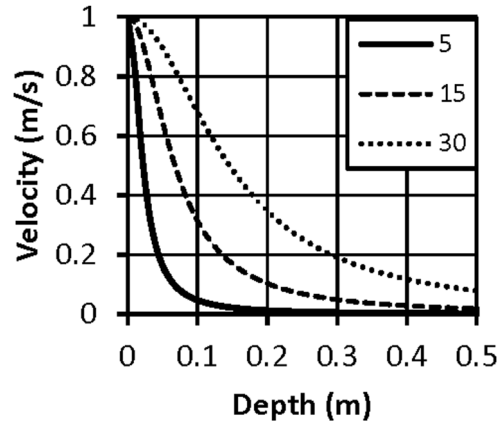


Fig. 5 Velocity of the wedge as a function of the penetration depth for three values of the deadrise angle $\beta = 5$ deg, 15 deg, 30 deg when $V_o = 1$ m/s and $M = 100$ kg

Figure 5 shows that the velocity drops quickly and that, as β becomes smaller, the velocity drops more abruptly. From Eq. (2.15), the acceleration is determined as follows:

$$\ddot{\zeta} = \frac{d}{dm} \left(\frac{MV_o}{M+m} \right) \cdot \frac{dm}{d\zeta} \cdot \frac{d\zeta}{dt} = - \frac{MV_o}{(M+m)^2} \cdot \frac{dm}{d\zeta} \cdot \frac{d\zeta}{dt} = - \frac{\dot{\zeta}^3}{MV_o} \frac{dm}{d\zeta} \quad (3.4)$$

For a wedge, m is given by Eq. (3.2), so

$$\ddot{\zeta} = - \frac{\pi \rho \gamma^2}{MV_o \tan^2 \beta} \zeta \dot{\zeta}^3 \quad (3.5)$$

The hydrodynamic force F is equal to the product of the mass M of the wedge times the acceleration

$$F = -M\ddot{\zeta} = \frac{\pi \rho \gamma^2}{V_o \tan^2 \beta} \zeta \dot{\zeta}^3 \quad (3.6)$$

Using Eq. (3.3), the force F can be written in terms of ζ only. Setting the derivative of F with respect to ζ to zero, we find that F reaches its maximum when the penetration depth is

$$\zeta^* = \sqrt{\frac{2M}{5\pi\rho\gamma^2}} \tan \beta \quad (3.7)$$

Substituting into Eq. (3.3), we find that the velocity at that time is

$$\dot{\zeta}^* = \frac{5}{6} V_o \quad (3.8)$$

From Eqs. (3.6)–(3.8), the maximum force is

$$F^* = \left(\frac{5}{6} \right)^3 \frac{V_o^2}{\tan \beta} \sqrt{\frac{2\pi}{5}} \rho M \gamma^2 \quad (3.9)$$

Using the method of separation of variables, Eq. (3.3) is written as

$$\left[1 + \frac{\pi}{2} \rho \frac{\gamma^2 \zeta^2}{M \tan^2 \beta} \right] d\zeta = V_o dt \quad (3.10)$$

and, after integration, the penetration depth is related to time implicitly by

$$t = \left[\zeta + \frac{\pi}{6} \rho \frac{\gamma^2 \zeta^3}{M \tan^2 \beta} \right] / V_o \quad (3.11)$$

Using Eq. (3.7), we find that the hydrodynamic force reaches its maximum for

$$t^* = \frac{16 \zeta^*}{15 V_o} = \frac{16}{15} \sqrt{\frac{2M}{5\pi\rho\gamma^2}} \frac{\tan\beta}{V_o} \quad (3.12)$$

Using Eqs. (3.9) and (3.12), we recover the result given by Carcaterra and Ciappi [36],

$$F^* t^* = \frac{20}{81} M V_o \quad (3.13)$$

During the penetration, water piles up near that point (Fig. 4) and Wagner (see Payne [37]) showed that the contour of the free surface is defined by $\eta(x) = (2\beta/\pi)x \sin^{-1}(c/x)$. As a result, the depth of the wedge immersed in water is $\pi\zeta/2$ and the radius of the added mass is $r' = \pi r/2$. Then, the parameter γ introduced earlier becomes $\gamma = \pi/2$. Different approaches have been proposed for calculating the added mass [38], resulting in different values of γ . It was also found [39,40] that γ is a function of the deadrise angle. Mei et al. [41] discussed how γ varies with the deadrise angle β . Various approaches have been used and there is agreement that $\gamma = 1$ when $\beta = \pi/2$ and $\gamma = \pi/2$ when $\beta = 0$. Pierson (see [41]) assumed that γ varies linearly between those two data points, but the numerical results of Mei et al. [41] show that this approximation leads to significant errors for intermediate values of the deadrise angle.

Figure 6 shows that the predictions from Eqs. (3.9), (3.12), and (3.13) are in good agreement with the experimental results from Carcaterra and Ciappi [36]. From Eq. (3.6), it can be shown that, if the velocity of the wedge is assumed to remain constant, the force F is given by

$$F = \frac{\pi\rho\gamma^2}{\tan^2\beta} V_o^3 t \quad (3.14)$$

This formula, cited in several publications (e.g., Ref. [42]), shows which parameters govern the rising portion of the force-time curve, but it does not predict the maximum force or when this maximum occurs. Skalak and Feit [43] and Feit et al. [44] presented a more general approach, including symmetric and asymmetric impact, supersonic impacts, and the effects of elasticity, which were found to be secondary.

3.2 Wagner's Approach for a Wedge. For a wedge, the length of the plate in Wagner's model (Fig. 7) is $r = (\pi/2)(\zeta/\tan\beta)$ when $\gamma = \pi/2$ in Eq. (3.2). Substitution into Eq. (2.33) gives

$$\frac{p}{\rho} = \dot{V} \sqrt{r^2 - x^2} + \frac{\pi}{2} \frac{V^2 r}{\tan\beta \sqrt{r^2 - x^2}} - \frac{1}{2} \frac{V^2 x^2}{r^2 - x^2} \quad (3.15)$$

where $V = \dot{\zeta}$. Often the acceleration of the wedge remains small and the first term on the right-hand side can be neglected. A number of studies assume a constant velocity V . Extensions of Wagner's method to three dimensions were proposed by several authors (e.g., Ref. [45]).

3.3 Complicating Factors. Several complicating factors should be considered in the study of the water entry of a rigid body:

- (1) the rise of the water level and the formation of a jet near the intersection point
- (2) air cushioning for blunt bodies
- (3) asymmetric and oblique impacts
- (4) acoustic effects

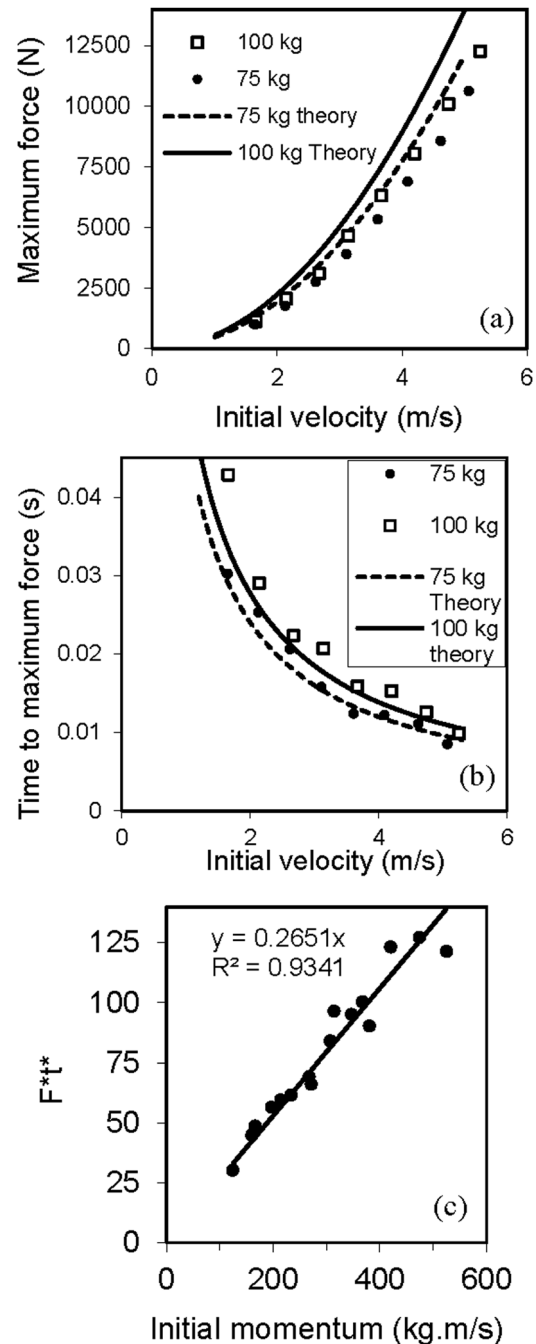


Fig. 6 Experimental results and theoretical predictions for the example of Carcaterra and Ciappi [36] ((a) maximum force versus initial velocity, (b) time to maximum force versus initial velocity, (c) product $F^* t^*$ in Eq. (3.13) versus initial momentum)

3.3.1 Jet Formation. During the initial stage of the impact of a body into a fluid, the fluid region is divided into three parts: (1) the bulk of the fluid, (2) the jet root region, and (3) the jet region. This is illustrated by the photograph from Greenhow and Lin [46] (Fig. 8) and the sketch in Fig. 9. Wagner's method ignores the jet root region, where the water piles up at the intersection between the body and the initial free surface of the water. It also ignores the jet region. As a result, the velocity of the fluid and the pressure become singular in the jet root region. The pressure can also become negative, which is not physically possible. In order to obtain a more accurate solution and fully account for the energy transferred from the body to the fluid, all three regions should be modeled. In that case, the root region is often called the inner flow

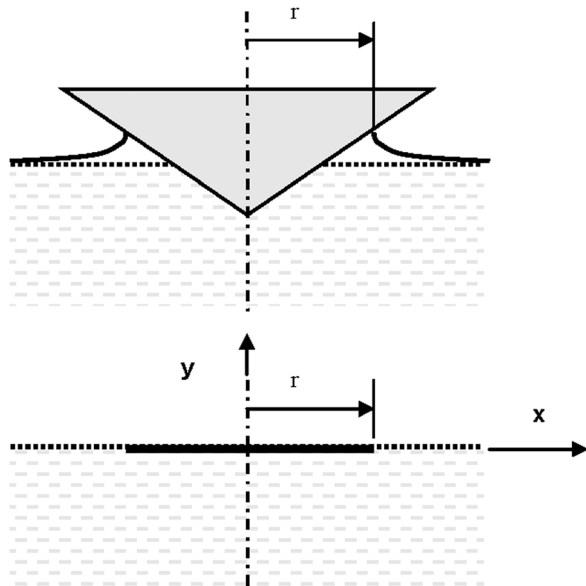


Fig. 7 Wagner's model (top: wedge entering the water; bottom: Wagner's expanding plate model)

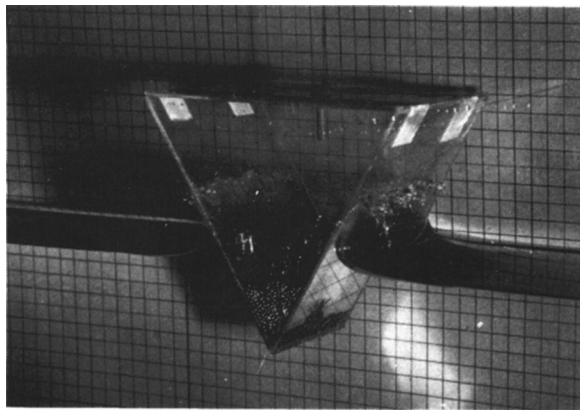


Fig. 8 Water entry of a wedge (Greenhow [52], reproduced with permission from Elsevier)

region and the bulk of the fluid is called the outer flow region. Using the asymptotic expansion method, separate approximations are made in each region and are then matched at the boundaries between the various regions.

Howison et al. [47] studied a class of two-dimensional bodies, including wedges with very small deadrise angles, so that the jet generated remains attached to the impacting body (Fig. 9). The impact is also assumed to be so rapid that gravity can be neglected. Explicit solutions are provided for the pressure distribution, and it was shown that the jet is wedge-shaped. The free surface of the jet makes a constant angle Γ with the edge of the body, and Mackie [48] has shown that this angle is related to the deadrise angle by $\cot \Gamma + \Gamma = \beta$. This problem was also studied by Cointe [49,50], who considered a wedge with a deadrise angle close to 90 deg using the same approach. Scolan and Korobkin [51] showed that the main part of the energy is transmitted to the jet if the body velocity decreases after the impact, which is the case for the free fall of a body onto the liquid free surface.

Flow separation is difficult to predict. Keeping the flow attached to the body in the analysis leads to the prediction of negative pressure on the surface of the body, as noted by Greenhow [52]. Fontaine and Cointe [53] presented an asymptotic solution with a rectilinear jet, making an angle θ with the wedge and, as the deadrise angle goes to zero, it is estimated that $\theta \approx \tan^2 \beta / (2\pi)$.

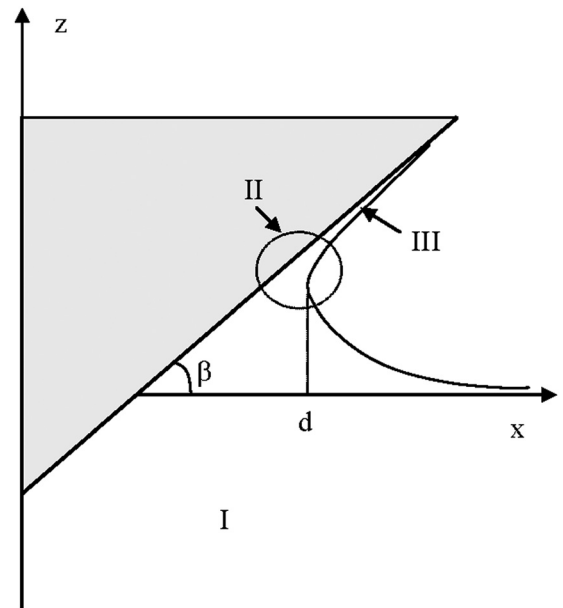


Fig. 9 Water entry of a wedge with small deadrise angle showing the three flow regions (I: outer flow region; II: inner flow region; III: jet flow region)

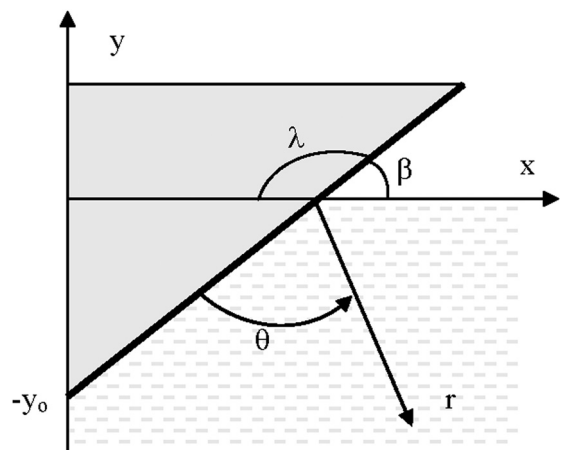


Fig. 10 Local coordinates in the inner region

Asymptotic solutions were presented by Iafrati and Korobkin [54–57]. In a small vicinity of the intersection point $(x_0, 0)$, where $x_0 = y_0 / \tan \beta$, the velocity potential behaves like

$$\phi \approx -Ar^{\sigma_0} \cos(\sigma_0 \theta) \quad (3.16)$$

where $\sigma_0 = \pi / (2\lambda)$, $\lambda = \pi - \beta$, and r and θ are the cylindrical coordinates with the origin at the intersection point (Fig. 10). Korobkin and Iafrati [58] considered the effect of fluid compressibility in an asymptotic analysis to obtain both the shape and the length of the spray jet.

Several authors [40,59–62] used what is called a similarity approach to solve the water-entry problem under a constant velocity, realizing that the flow is geometrically self-similar when the effect of gravity is neglected. Introducing the nondimensional variables $\tilde{x} = x/V_0 t$ and $\tilde{y} = y/V_0 t$, the velocity potential can be written as $\phi(x, y, t) = V_0^2 t \phi(\tilde{x}, \tilde{y})$ so that the time variable is eliminated from the formulation of the problem.

Formulating the problem and solving the resulting equations is rather challenging, so that the use of numerical methods is being

investigated. In Battistin and Iafrati [63,64], a special approach is developed to describe the flow and the pressure field inside the thin jet layer, while the rest of the fluid is modeled using the boundary element method.

For the water impact of cylinders and spheres, using a nonlinear boundary-element method with the jet domain cut off from the computational domain, Battistin and Iafrati [65] evaluated the kinetic energy in the jet (E_J) and in the bulk of the fluid (E_B) and found that the ratio $E_J / (E_J + E_B)$ can be significant (0.463 when $\beta = 10$ deg, for example).

3.3.2 Trapped Air. For a body with a flat bottom or a very small deadrise angle impacting the surface of the water, Wagner's approach predicts that the pressure becomes infinite. If the pressure becomes high, considering the compressibility of the water, the speed of sound in the water c is approximately 1500 m/s and the pressure is the product of the impedance ρc and the impact velocity V_o

$$p_{\max} = \rho c V_o \quad (3.17)$$

Several investigators [40,66–69] showed that this equation overestimates the pressure and that the entrapped air action and its cushioning effects cannot be neglected. Chuang [70] conducted experiments for small deadrise angles (0 deg to 15 deg) and showed that the maximum pressure is proportional to V_o^2 instead of cV_o in the acoustic approximation (Eq. (3.17)). The first positive pressure pulse was shown to occur when air is trapped momentarily between the falling body and the water surface. Verhagen [71] presented an analytical model considering the falling flat-bottomed body, a layer of air, and the water. Experiments were conducted and, with the help of a high-speed camera, it was shown that water elevation is noticeable near the edges of the plate just before it touches the water (Fig. 11). Once contact with the water is made near the edges, the air layer breaks up into bubbles starting from the edges and moving towards the center of the plate. With increasing time, the bubbles move slowly away from under the plate. Similar experimental observations were reported later by Oh et al. [72], and numerical approaches for this problem were developed by Iwanowski and Yao [73] and Koehler [74]. Huang and Hsiung [75] included the deformation of the bottom of the body (elastic plate) and considered the flow of air to be two-dimensional instead of one-dimensional in Ref. [17]. A numerical study [76] of the impact of a rectangular rigid body, including the flow of air between the body and the surface of the water, showed that the impact starts well before the body comes in contact with water, and the force reaches a maximum, while a thin layer of air is still present. After that peak, contact with the water is made and the simulation shows the presence of air pockets under the body. Another numerical analysis [77] indicates that, for very small angles (0 to 3 degrees), there may be four types of behavior as the momentum per unit area increases: (1) a bubble forms and pressure peaks before impact; (2) no bubble forms and pressure peaks before impact; (3) no bubble forms and pressure peaks after impact; and (4) bubble forms and pressure peaks after impact. Additional experimental results are provided by Lin and Shieh [78].

3.3.3 Oblique or Asymmetric Water Impacts of a Wedge. Oblique or asymmetric water impacts of a wedge occur when the plane of symmetry of the wedge is not perpendicular to the

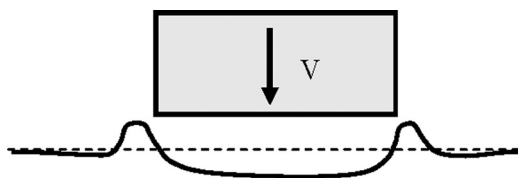


Fig. 11 Air cushioning of flat bottom body prior to water entry

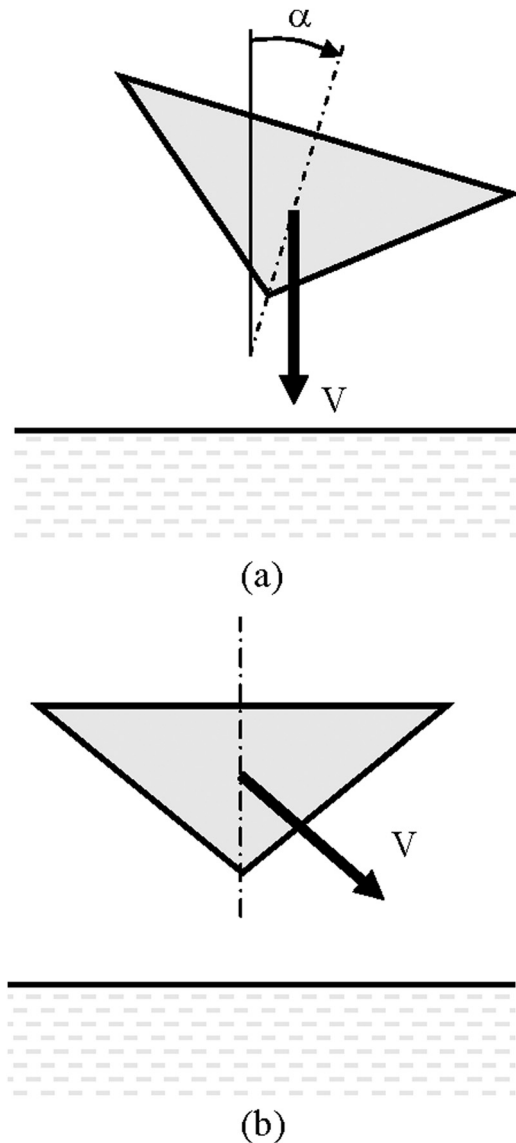


Fig. 12 (a) Asymmetric water impact of a wedge; (b) oblique water impact of a wedge

surface of the water or when the initial velocity is not normal to the water surface (Fig. 12). Garabedian [79] appears to be the first to consider the oblique water entry of a wedge. Chekin [80] distinguished three cases: (1) the water flows past the wedge without breaking away; (2) one face is not wetted (semi-infinite plate); and (3) a cavity forms on one face. Chekin [81] proposed a criterion for nonseparated flow in the form of a relation between three angles, defining the angle of the wedge, the direction of the entry velocity, and the angle of attack. Further analyses of this problem were presented by several authors [82–87]. Hua et al. [88] used the extended Wagner approach of Toyama [89] to calculate the pressure under an asymmetric wedge and used these results to determine the dynamic deflections of plate structure under that loading. Riccardi and Iafrati [90,91] studied water impacts on floating asymmetric wedges.

3.3.4 Water Compressibility. Most analyses consider the fluid to be incompressible. However, there are cases for which the compressibility of the fluid might be significant. Korobkin and Pukhnachov [92] presented a review of earlier work on the initial stages of solid-liquid impacts. In this section, we briefly consider the subsequent literature.

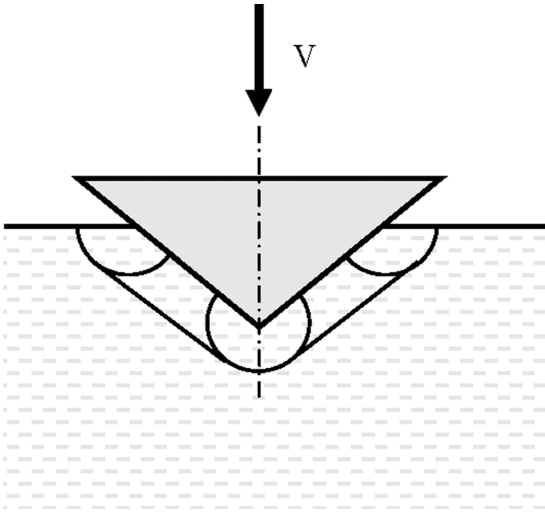


Fig. 13 Floating wedge subjected to an impulsive start

Considering a floating wedge subjected to a sudden impulse (Fig. 13), as in Korobkin and Iafrati [93], following Huygens approach, we see that compressive waves propagate through the water at a speed c of nearly 1500 m/s. As the wedge moves down by a distance ζ , the horizontal velocity of the intersection point of the wedge with the initial free surface is $\dot{r} = \zeta / \tan \beta$. If the penetration velocity is of the order of a few meters per seconds, as it is in the case for hull slamming problems, $\dot{r}$ never exceeds c and the impact is subsonic. In the case of a bullet fired into the water with a speed of 3000 ft/s, $\dot{r}$ is larger than 1500 m/s when $\beta \leq 31$ deg. Then, the impact is supersonic in the sense that the intersection point moves faster than the speed of sound and prevents the shock wave from reaching the free surface. However, the projectile slows down under the action of the hydrodynamic force and the impact will quickly become subsonic.

In practice, for hull slamming problems, the velocity $\dot{\zeta}$ is rather small compared to c ($\dot{\zeta}/c \ll 1$); the velocity $\dot{r}$ is larger than the speed of sound in water ($\dot{r} = \dot{\zeta}/\tan \beta > c$) only if the deadrise angle β is almost zero (flat bottom). So the motion of the wedge is not supersonic, but, for a short time ($t < t^*$), a portion of the wetted edge is subjected to the compressive wave with a pressure of the order of $\rho c V$. This compressive phase lasts until the compressive wave centered at the vertex of the wedge meets the relief wave coming from the intersection point. This occurs at time $t^* = L/(2c)$, where L is the length of the wetted portion of the wedge. A numerical analysis of the jetting flow generated during the early stage after the impulsive vertical start of a floating wedge [93] showed that the velocity of the jet tip is controlled by the liquid compressibility and, therefore, cannot be evaluated within the incompressible liquid model.

Campana et al. [94] showed that the effects of water compressibility are negligible for wedges and horizontal cylinders entering the water surface. The maximum impact force occurs during the incompressible phase of the motion.

Acoustic effects during water impacts of blunt bodies have been investigated by Korobkin and Iafrati [93,95–98]. Korobkin [98] studied shallow-water impacts of a boxlike structure and distinguished: (1) the early time response with a duration of the order of h/c when compressibility effects are of major importance; (2) the second stage with a duration of the order $L = c$, during which the outflow from beneath the entering body will develop; and (3) the third stage, the duration of which is of the order h/V_0 , the outflow of liquid from beneath the body has already developed, and the flow is unsteady and nonlinear. In the first stage, the outflow of liquid from beneath the body is not developed and the horizontal component of the liquid velocity is zero, except for a small

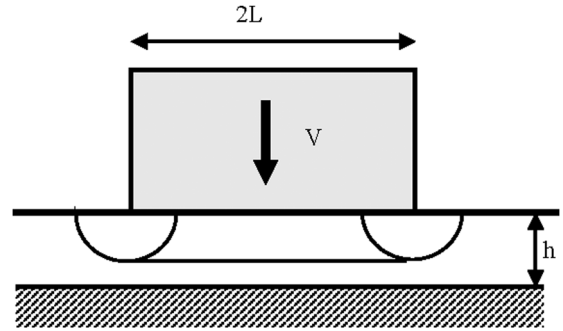


Fig. 14 Impact of flat-bottom body on shallow water. Compressive wave propagating downwards through the water.

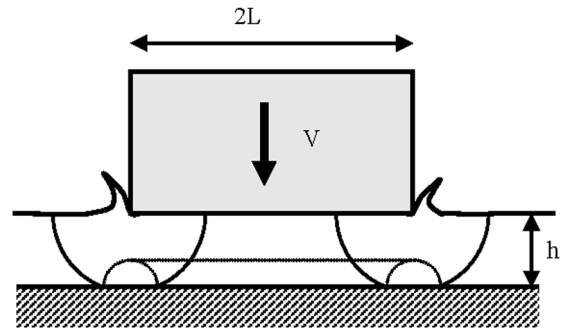


Fig. 15 Impact of flat-bottom body on shallow water. The shock wave reflected from the bottom and the outflow of the liquid from beneath the body is still not developed (adapted from Korobkin [98]).

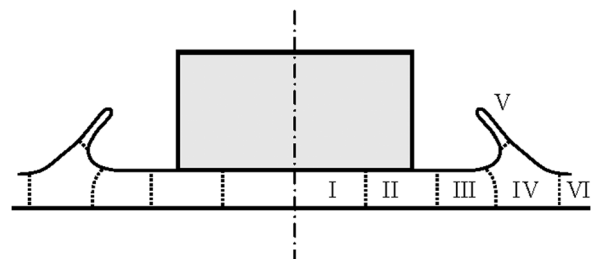


Fig. 16 Asymptotic expansion model for the third stage (adapted from Korobkin [98])

region close to the edges of the body (Fig. 14). Figure 15 shows the various wave fronts after reflection of the acoustic wave from the bottom. An asymptotic expansion model for the third stage of the impact was developed by dividing the fluid domain into six different regions (Fig. 16). The asymptotic expansion method will be discussed further in connection with the water impact of cylinders and plates.

3.3.5 Elastic Wedge Impacts. Several authors considered the water impact of a two degree of freedom system consisting of a wedge, a spring, and another mass (Fig. 17). Miller and Merten [99] used this type of model to analyze the water landing of a seaplane. The upper mass and the spring are used to simulate the relative motion of the wings of the airplane. This model was used by Carcaterra and Ciappi [36] and Iafrati et al. [42], with the upper mass and the spring representing an elastic system carried on board a ship. Bogaert and Kaminski [100] used this two degree of freedom model to study the water impact of flexible cones. Carcaterra and Ciappi [101] also considered the impact of a wedge carrying a damped single degree of freedom (SDOF) system in a configuration similar to that in Fig. 17 with a dashpot in parallel

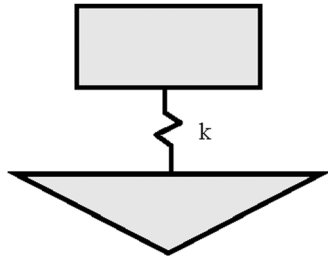


Fig. 17 Two degree of freedom model for water impact

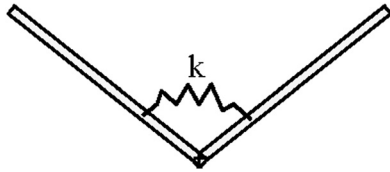


Fig. 18 Elastic wedge model: two beams pinned at the vertex and connected by a rotational spring

with the spring. A three degree of freedom system for water entry was used by Gollwitzer and Petersen [102] to account for the motion of the boat, the deck on which the human is seated, and the human.

Several investigators developed simple models for flexible wedges in which the sides are modeled as elastic beams. Carcaterra and Ciappi [101] considered the water entry problem of a wedge consisting of two rigid beams connected by a pin at the vertex and a rotational spring (Fig. 18). Carcaterra et al. [103] considered the water entry of a wedge in which the two sides are flexible and are modeled as simply supported plates. Similarly, Khabakhpasheva and Korobkin [104] studied the water impact of wedges where the two sides in contact with water are simply supported Euler beams. The Wagner approach used to analyze the water entry problem of a rigid wedge is coupled with the dynamic beam deflection problem formulated in terms of the normal modes of the beam. A similar study was published by Datta et al. [105] and Maki et al. [106]. Korobkin et al. [107] analyzed the same problem by coupling Wagner solution and a finite element model of the beams. Luo et al. [108] used a finite element approach to simulate the dynamic response of flat, stiffened panels under slamming loads and discussed the role of hydroelasticity, including air cushioning, impact velocity, and panel stiffness. Panciroli et al. [109] conducted experiments with flexible wedges made of steel or composite materials and used the smoothed particle hydrodynamics (SPH) method to model the water entry under various conditions. Particular attention was given to the onset fluid structure interaction. It was found that, in order to induce fluid-structure interaction, the duration of the impact must be close to the period of the first mode of the side panels. In that case, the evolution of the acceleration versus time is similar to that of a rigid wedge until a maximum is reached. Beyond that, oscillations are superposed on the decaying portion of the acceleration versus time curve. A numerical simulation indicated that the oscillations are due to successive separations and impacts between the side panels and the water.

4 Water Entry of Rigid Bodies With Other Shapes

In addition to the water entry of rigid wedges, previous investigators have considered the water entry of rigid bodies of various shapes: spheres, cones, cylinders, flat plates, and a number of other shapes. This section provides an introduction to the literature in this area and discusses several new topics, such as the ricochet of spheres off the water surface, the skipping of stones, and

the method of asymptotic expansion for obtaining analytical solutions for early times.

4.1 Water Entry of Spheres. The water entry of a sphere has been studied for many years, starting with the work of Worthington [5], who examined the water entry of a sphere in detail and first described the formation of a splash, the formation and the closure of a cavity, and the formation of a jet. In 1975, May [110] reviewed prior work on the subject and divided the water entry into several phases:

- (1) the shock-wave phase, during which a small region near the wetted surface of the projectile is subjected to a shock wave. This phase is of very short duration and of little practical importance.
- (2) the flow-forming phase, during which the projectile establishes a pattern of flow in the water that was originally at rest
- (3) the open cavity phase, in which the projectile is generating an open cavity that keeps growing as it travels
- (4) a closed-cavity phase, in which the cavity closes and moves with the projectile
- (5) a collapsing cavity phase
- (6) a fully wetted phase, where the water is in contact with the complete surface of the projectile (sometimes called missile in some publications)

This subsection deals with the shock wave phase, the cavity forming phase, and the ricochet of spheres off of the surface of the water. The open cavity phase and the closed cavity phase will be discussed later in conjunction with the hydrodynamic ram problem.

4.1.1 Shock Wave Phase. When the sphere impacts the water surface, it generates a compression wave with a hemispherical wave front propagating at the speed of sound in water ($c=1500$ m/s). Considering the geometry of the problem (Fig. 19), ζ is the penetration depth relative to the initial water surface, OA is a radius of the sphere ($OA=R$), and the distance AB is called r . As the sphere moves down, we will show that, initially, $\dot{r}$, the horizontal velocity of the intersection point with the initial surface is larger than c , so the shock wave remains trapped under the projectile. As $\dot{r}$ becomes smaller than c , the shock wave phase ends and the cavity formation begins.

Since OAB is a right-angled triangle, $R^2 = r^2 + (R - \zeta)^2$ and $r = (2R\zeta - \zeta^2)^{1/2}$. The horizontal velocity of the intersection point is given by

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\zeta} \frac{d\zeta}{dt} = \frac{1 - \frac{\zeta}{R}}{\sqrt{2\frac{\zeta}{R} - \left(\frac{\zeta}{R}\right)^2}} \dot{\zeta} \quad (4.1)$$

The velocity $\dot{r}$ becomes infinite as ζ goes to zero, and it will remain higher than 1500 m/s, the speed of sound in water, until

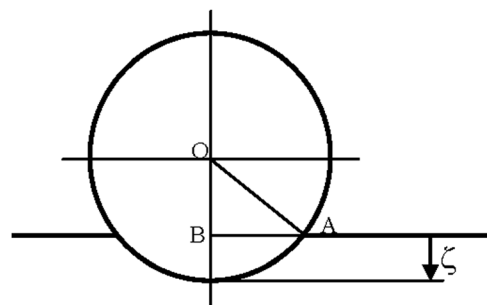


Fig. 19 Penetration of the initial water level by a sphere

Table 1 Nondimensional penetration depth $\zeta_{\max}/R$ when $\dot{r} = 1500$ m/s for several values of the initial velocity of the sphere

V_o (m/s)	1	2	5	10
$\zeta_{\max}/R$	2.22×10^{-7}	8.88×10^{-7}	5.555×10^{-6}	2.222×10^{-5}

the penetration depth ζ reaches the value $\zeta_{\max}$. Table 1 shows that, for small initial velocities, $\dot{r} > 1500$ m/s only when ζ is very small compared to R , the radius of the sphere. Since the duration of this supersonic radius expansion phase is small, the velocity of the sphere remains nearly constant ($\dot{\zeta} \approx V_o$). Then, Eq. (4.1) gives the approximate relation

$$\zeta_{\max}/R \approx \frac{1}{2} (V_o/1500)^2 \quad (4.2)$$

When the velocity is only a few meters per seconds, the shock wave phase is so short that it is usually neglected in the analysis of water entry. On the other hand, if the velocity is up to 1463 m/s, as in McMillen [111], the pressure is 2.195 MPa and the impact phase of the water entry lasts until $\zeta_{\max}/R = 0.2481$. In that case, the shock wave phase is very significant.

4.1.2 Motion of the Sphere in the Cavity-Forming Phase. With the momentum approach, the velocity during impact is given by Eq. (2.14). The acceleration

$$\ddot{\zeta} = \frac{d}{dm} \left(\frac{MV_o}{M+m} \right) \cdot \frac{dm}{d\zeta} \cdot \frac{d\zeta}{dt} \quad (4.3)$$

which can be written as

$$\ddot{\zeta} = - \frac{(MV_o)^2}{(M+m)^3} \cdot \frac{dm}{d\zeta} \quad \text{or} \quad \ddot{\zeta} = - \frac{\dot{\zeta}^3}{MV_o} \frac{dm}{d\zeta} \quad (4.4)$$

Following the approach used for the wedge, several authors, including Richardson [112], take the added mass of the fluid to be the mass of a hemisphere of radius r

$$m = \frac{2\pi}{3} r^3 \rho = \frac{2\pi}{3} \rho (2R\zeta - \zeta^2)^{3/2} \quad (4.5)$$

$$\frac{dm}{d\zeta} = 2\pi\rho [2R\zeta - \zeta^2]^{1/2} (R - \zeta) \quad (4.6)$$

The mass of the sphere being $M = (4\pi/3)\rho_s R^3$, the ratio of the two masses becomes

$$\frac{m}{M} = \frac{1}{2} \frac{\rho}{\rho_s} \frac{[2R\zeta - \zeta^2]^{3/2}}{R^3} \quad (4.7)$$

Using Eqs. (4.6) and (4.7), the acceleration (second part of Eq. (4.4)) becomes

$$\ddot{\zeta} = - \frac{2\pi\rho V_o^2}{M} \frac{[2R\zeta - \zeta^2]^{1/2} (R - \zeta)}{\left[1 + \frac{1}{2} \frac{\rho}{\rho_s} \frac{[2R\zeta - \zeta^2]^{3/2}}{R^3} \right]^3} \quad (4.8)$$

Since $F = -M\ddot{\zeta}$, Eq. (4.8) gives the hydrodynamic force as a function of the penetration depth ζ , which can be rewritten in terms of the nondimensional depth $\bar{\zeta} = \zeta/R$ as

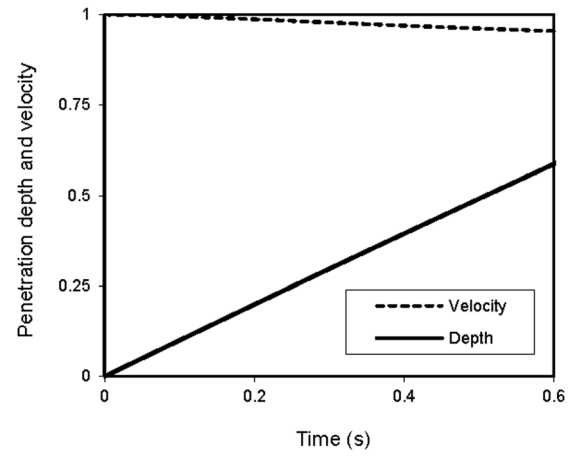


Fig. 20 Water entry of a steel sphere

$$F = 2\pi\rho V_o^2 R^2 \frac{[2\bar{\zeta} - \bar{\zeta}^2]^{1/2} (1 - \bar{\zeta})}{\left[1 + \frac{1}{2} \frac{\rho}{\rho_s} [2\bar{\zeta} - \bar{\zeta}^2]^{3/2} \right]^3} \quad (4.9)$$

Combining the first part of Eqs. (4.4) and (4.6), the hydrodynamic force can also be written as

$$F = \frac{M^3 V_o^2}{(M+m)^3} 2\pi\rho [2R\zeta - \zeta^2]^{1/2} (R - \zeta) \quad (4.10)$$

To understand this phase of the motion, we consider three examples. The first one is a steel sphere with $\rho_s = 7850$ kg/m³ and $\rho/\rho_s = 0.1274$. Figure 20 shows that the velocity of the projectile is only slightly affected by the impact with the water surface. Using Eq. (2.16), we find that, when $\zeta/R = 0.5$, the velocity of the sphere is reduced to 96% of its initial value ($\dot{\zeta}/R = 0.9603$). The second example (Fig. 21) is that of a golf ball with a mass of 45.93 g and a diameter of 42.67 mm. The effective density of that ball is 1128.9 kg/m³, so that $\rho/\rho_s = 0.8858$. For this golf ball, when $\zeta/R = 0.5$, the velocity of the sphere is reduced to 78% of its initial value ($\dot{\zeta}/R = 0.7766$). The third example is that of a ping pong ball with a 40-mm diameter and a mass of 2.7 g. The effective density is 80.57 kg/m³ and $\rho/\rho_s = 12.41$. In that case, the water entry reduces the velocity very significantly (Fig. 22). For this ball, when $\bar{\zeta} = 0.5$, the velocity of the sphere is reduced to 20% of its initial value ($\dot{\zeta}/R = 0.1988$).

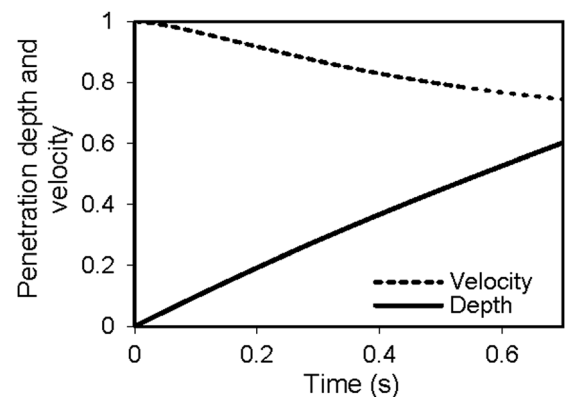


Fig. 21 Water entry of a golf ball

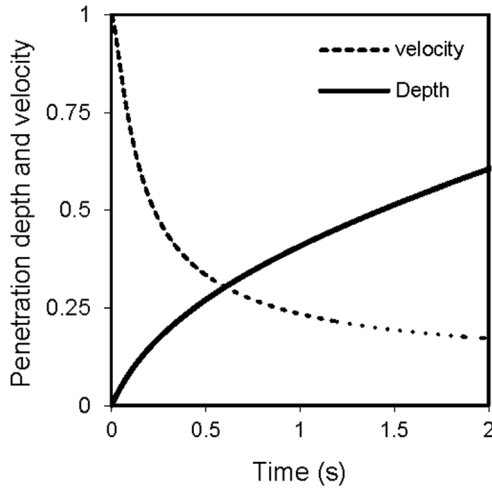


Fig. 22 Water entry of a ping pong ball

Since m depends on the penetration depth, it is not easy to determine when the force F reaches a maximum. However, if $m \ll M$, Eq. (4.10) simplifies to

$$\frac{F}{\rho V_0^2} \approx 2\pi[2R\zeta - \zeta^2]^{1/2}(R - \zeta) \quad (4.11)$$

and then, by taking the derivative of the right-hand side with respect to ζ , it can be shown that a maximum is reached when the penetration depth is

$$\zeta^* = \left(1 - \frac{1}{\sqrt{2}}\right)R \quad (4.12)$$

which corresponds to the situation shown in Fig. 19, where the angle AOB = 45 deg. The maximum force is then

$$F^* = \pi\rho R^2 V_0^2 \quad (4.13)$$

Equations (4.12) and (4.13) provide an estimate of the maximum force and the corresponding penetration depth when $m \ll M$. Equation (4.10) obtained by Richardson [112] is cited by several authors. Considering three examples: a steel ball ($\rho_s = 7850 \text{ kg/m}^3$), an aluminum ball, and a ping pong ball ($\rho_s = 80.57 \text{ kg/m}^3$) impacting water ($\rho_s = 80.57 \text{ kg/m}^3$), the nondimensional force $\tilde{F} = F/(\pi\rho R^2 V_0^2)$ is plotted versus the non-dimensional penetration depth ζ/R . The approximate equations (Eqs. (4.12) and (4.13)) predict that the force reaches its maximum when $\zeta/R = 0.29289$ and that $\tilde{F} = 1$ then. Figure 23 shows that the maximum occurs for smaller values of ζ/R and that the maximum value of $\tilde{F}$ can be significantly less than one.

Initially, $m \ll M$ and $\zeta \ll R$, so the hydrodynamic force can be approximated by

$$F \approx 2^{3/2}\pi\rho V_0^{5/2} t^{1/2} R^{3/2} \quad (4.14)$$

In an example from Hoover and Dawson [113], a 2-in-diameter aluminum sphere enters the surface of the water at 28 ft/s. The approach described here predicts a maximum acceleration of 465 m/s^2 after $715 \mu\text{s}$ and, at that time, $\zeta = 0.237$. These results are in excellent agreement with those in Ref. [113]. In the calculations, the density of aluminum was taken to be 2700 kg/m^3 and the parameter was taken to be 0.64. Hoover and Dawson also pointed out that the first natural frequencies of a sphere are given by

$$f = \frac{0.3c_s}{R(1 + \nu_s)^{1/2}} \quad (4.15)$$

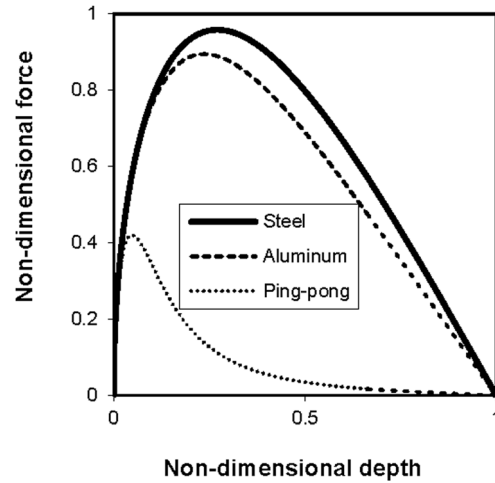


Fig. 23 Nondimensional force $\tilde{F} = F/(\pi\rho R^2 V_0^2)$ versus the non-dimensional penetration depth ζ/R for three spheres with different specific densities: steel ball, aluminum ball, ping pong ball

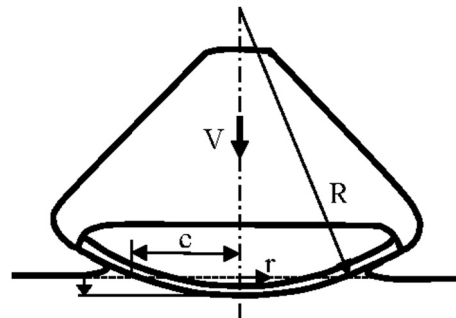


Fig. 24 Water entry of Apollo capsule (Cappelli et al. [116])

where c_s is the speed of sound in the sphere, R is the radius, and ν_s is Poisson's ratio. This equation predicts a natural frequency of 52,300 Hz or a period of $19 \mu\text{s}$. This corresponds to the period of the high frequency oscillations in the pressure measurements.

Cappelli and Wilkinson [114] studied the vertical impact of the Apollo capsule into water. The bottom of that vehicle has a spherical shape (Fig. 24) with a radius of 175 in (4.445 m). The other relevant parameters of the problem are the weight of the vehicle (10,000 lb, corresponding to a mass of 4530 kg), the density of water (62.5 lb/ft^3 or 1048 kg/m^3), and the impact velocity. In that reference, instead of using Eq. (4.5), the added mass of the fluid is calculated using

$$m = \frac{4}{3}r^3\rho = \frac{4}{3}\rho(2R\zeta - \zeta^2)^{3/2} \quad (4.16)$$

This expression is based on the added mass for a circular plate of radius r moving through a fluid, taken from Lamb [115], and is used by a number of other authors (e.g., Hoover and Dawson [113]). This value of m (Eq. (4.16)) is equal to $(2/\pi)((2\pi/3)r^3\rho) = 0.64((2\pi/3)r^3\rho)$, and it is often said to be 64% of the mass of the hemisphere of radius r . The ratio of this added mass and the mass M of the vehicle is

$$\frac{m}{M} = \frac{1}{\pi} \frac{\rho}{\rho_s} \frac{[2R\zeta - \zeta^2]^{3/2}}{R^3} = \frac{4}{3} \frac{\rho(2R\zeta - \zeta^2)^{3/2}}{M} \quad (4.17)$$

$$\frac{dm}{d\zeta} = 4\rho[2R\zeta - \zeta^2]^{1/2}(R - \zeta) \quad (4.18)$$

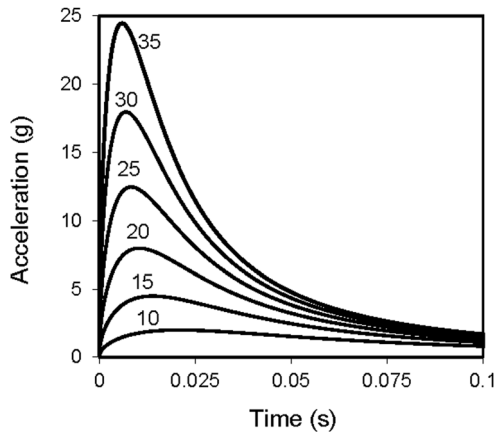


Fig. 25 Acceleration versus time for vertical impact of Apollo capsule

Substituting into Eq. (4.4), the hydrodynamic force becomes

$$F = -M\ddot{\zeta} = \frac{\dot{\zeta}^3}{V_0} 4\rho [2R\dot{\zeta} - \dot{\zeta}^2]^{1/2} (R - \dot{\zeta}) \quad (4.19)$$

where $\dot{\zeta} = V_0 / (1 + (m/M))$. Figure 25 shows how the acceleration varies with time for several values of the initial velocity: 35, 30, 25, 20, and 15 ft/s (1 ft/s = 0.3048 m/s). The acceleration increases very rapidly, reaches a maximum, and then decreases more gradually. The maximum occurs at a later time, when the impact velocity is decreased. These results are in very good agreement with those presented in Fig. 2 of Cappelli and Wilkinson [114]. It must be noted that the present values of the accelerations are slightly lower than those given in that reference, because it is further assumed that the velocity of the capsule remained constant and substituted $\dot{\zeta} = V_0 t$ into Eq. (4.17). It can be seen, from Eq. (4.10), that the maximum force is directly proportional to the square of the initial velocity V_0 and that this maximum occurs for the same value of the penetration depth ζ for all values of V_0 . This latter point is illustrated in Fig. 26.

Cappelli and Wilkinson also plot pressure distributions on the wetted surface of the vehicle at different times. This was done using Wagner's approach. Since the velocity was assumed to remain constant, the first term in Eq. (2.23) was not included. This might be acceptable, since Fig. 3 of that reference gives results for times larger than 0.1 s. By then, the accelerations are very small (Fig. 25), but this term should be included for determining pres-

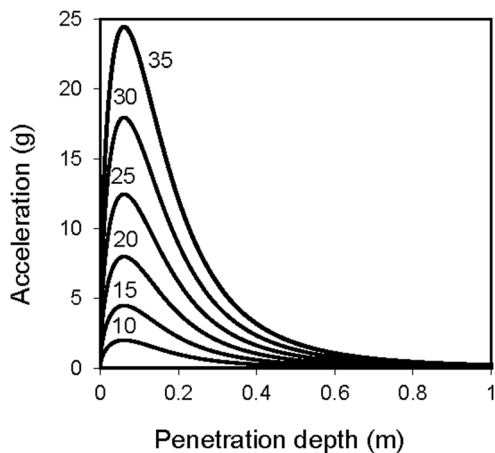


Fig. 26 Acceleration versus penetration depth for vertical impact of Apollo capsule

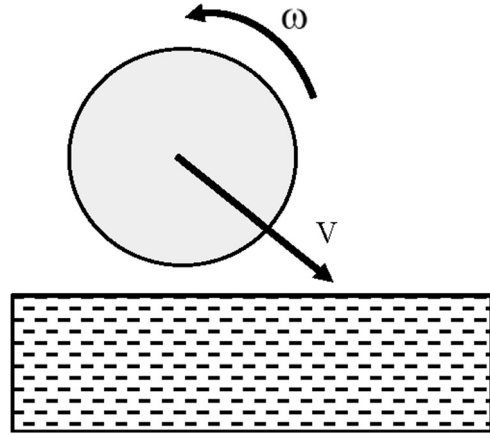


Fig. 27 Sphere impacting the surface of the water with an oblique linear velocity V and an angular velocity ω

sure distribution for earlier times. The last term in Eq. (2.23) was also neglected, because it leads to negative pressures near the edge. These pressure distributions are then used in a separate analysis to determine the dynamic deflections of the bottom portion of the vehicle, modeled as a spherical shell panel. In a subsequent study, Cappelli et al. [116] included the hydroelastic interaction between the shell and the fluid to account for pressure oscillations due to dynamic deflections of the shell during impact.

Nisewanger [117] measured the pressure distribution on a 12-inch-diameter sphere during vertical water entry at 23.5 fps versus time measurements at several points on the sphere. Plots of pressure versus angle from the stagnation point were obtained for various penetration depths, and the rise of the free surface of the water versus penetration was determined. These results provide insight into the mechanics of water entry by a sphere.

4.1.3 Ricochet of Spheres Off Water. In the most general case, spheres can impact the surface of the water at an angle θ with a linear velocity V and an angular velocity ω (Fig. 27). The sphere will be subjected to a force F that can be decomposed into a lift and a drag force (Fig. 28). As discussed by Richardson [112]

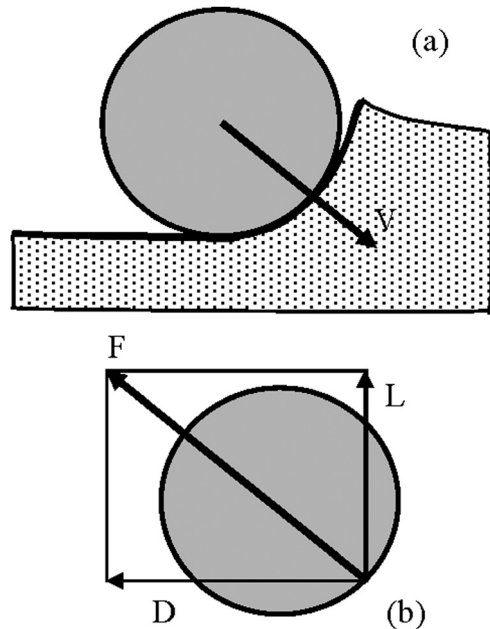


Fig. 28 Oblique impact of a sphere on the surface of a fluid. (a) Contact between the fluid and the water; (a) total force F and (b) lift and drag components.

and Johnson [118], the projectile will follow a curved path that will result in either bouncing off the surface of the water or sinking. Considering the water impact of nonspinning spheres, Soliman et al. [119] conducted tests that showed that ricochets are possible when the impact angle is less than the critical angle (in degrees) given by

$$\theta_c = 18/\sqrt{\rho_s/\rho} \quad (4.20)$$

where ρ_s is the density of the sphere and ρ is the density of water. This critical angle is 6.42 deg for steel spheres and 10.95 deg for aluminum spheres. When $\theta < \theta_c$, the velocity of the sphere must exceed a certain value for the sphere to bounce off the surface of the water. That velocity increases with θ and tends to infinity as $\theta \rightarrow \theta_c$. Equation (4.20) is based on an assumption on the pressure distribution on the wetted surface of the sphere made by Birkhoff. A different assumption made by Rayleigh leads to the same result [120]. These results are in excellent agreement with the previous experimental results of Richardson [112], who noted that, in 1883, de Jonquieres [121] concluded that “the product of the critical angle and the square root of the specific gravity is approximately constant.” Ricochets off sand were studied by Soliman et al. [119] and by Daneshi and Johnson [122] and Bai and Johnson [123].

Hutchings [120] showed that, for spinning spheres, the critical angle is given by

$$\theta_c = 18.7 \left(1 + 1.6 \frac{R\omega}{V} \right)^{1/2} / \sqrt{\rho_s/\rho} \quad (4.21)$$

where the angular velocity ω is taken to be positive for underspin, as shown in Fig. 27. Underspin does increase θ_c , while top spin results in a decrease. The ricochet of spinning spheres or cylinders was of great interest in the development of the Wallis bouncing bomb [124,125].

The minimum velocity for a glancing impact is given by

$$v_{\min} = \sqrt{gR} \left[\frac{20}{\pi} \sqrt{\rho_s/\rho} + 2.5 \right] \quad (4.22)$$

This expression is attributed to Miloh and Shukron [126] in de Podesta [125]. An analysis of the ricochet of spherical projectiles [127,128] concluded that Eq. (4.18) may be considered as an approximation only for infinitely large Froude numbers. The experiments of Truscott and Techet [129,130] showed that, for spinning spheres, spinning motion induces a lift force on the sphere and thus causes significant curvature in the trajectory. Shlien's experiments [131] showed that cavitating and spinning spheres can ricochet off the free surface of water at incident angles as large as 45 deg. Spinning spheres can ricochet backwards.

4.2 Water Entry of Cones. The mechanics of the water entry of rigid cones are considered using von Karman's approach and Wagners's approach, and we also discuss the oblique impact of a cone on the water surface.

4.2.1 Von Karman's Approach for the Water Entry of Cones. As discussed in Sec. 4.1, one approach for calculating the added mass of the fluid is to consider the added mass of a circular plate with a radius r equal to the radius of the cone at the intersection with the initial free surface (Fig. 29). Then, $m = \frac{4}{3}r^3\rho$, where r is related to the penetration depth and cone half angle by $r = \zeta \tan \beta$. Therefore,

$$m = \frac{4}{3} \rho (\zeta \tan \beta)^3 \quad (4.23)$$

This expression has been used by several authors, including Baldwin [132]. Using the conservation of momentum approach, the velocity of the cone is

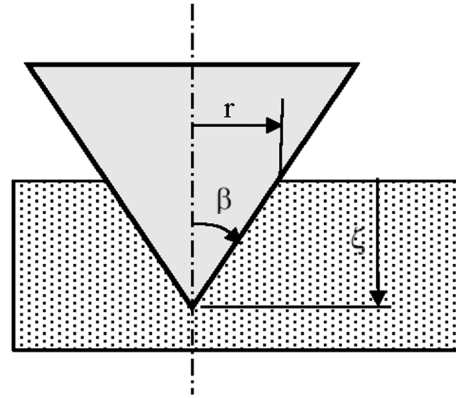


Fig. 29 Water entry of a cone

$$\dot{\zeta} = \frac{MV_o}{M+m} = \frac{MV_o}{M + \frac{4}{3}\rho\zeta^3 \tan^3 \beta} \quad (4.24)$$

and, from Eq. (4.4), the acceleration can be written as

$$\ddot{\zeta} = -\frac{\dot{\zeta}^3}{MV_o} \frac{dm}{d\zeta} = -\frac{(MV_o)^2}{\left(M + \frac{4}{3}\rho\zeta^3 \tan^3 \beta\right)^3} (4\rho\zeta^2 \tan^3 \beta) \quad (4.25)$$

The hydrodynamic force is given by

$$F = -M\ddot{\zeta} = \frac{4\rho\zeta^2 \tan^3 \beta}{\left(1 + \frac{4}{3M}\rho\zeta^3 \tan^3 \beta\right)^3} V_o^2 \quad (4.26)$$

Integrating Eq. (4.23) gives the following relationship between penetration depth and time:

$$t = \frac{\zeta}{V_o} \left(1 + \frac{1}{3M}\rho\zeta^3 \tan^3 \beta \right) \quad (4.27)$$

Taking the derivative with respect to ζ , we find that the acceleration reaches its maximum when the penetration depth becomes

$$\zeta^* = \frac{1}{\tan \beta} \left(\frac{3M}{14\rho} \right)^{1/3} \quad (4.28)$$

Substituting into Eqs. (4.23) and (4.24), we find that, at that instant, the added mass and the velocity are

$$m = \frac{4}{14}M, \quad \dot{\zeta}^* = \frac{7}{9}V_o \quad (4.29)$$

Using Eq. (4.25), the maximum force is found to be

$$F^* = 4 \left(\frac{7}{9} \right)^3 \rho^{1/3} M^{2/3} V_o^2 \tan \beta = 1.8820 \rho^{1/3} M^{2/3} V_o^2 \tan \beta \quad (4.30)$$

Using Eq. (4.26), the maximum force is found to occur at time t^* given by

$$\begin{aligned} t^* &= \frac{15}{14} \frac{\zeta^*}{V_o} = 5(3/14)^{4/3} (M/\rho)^{1/3} / (V_o \tan \beta) \\ &= 0.64115 (M/\rho)^{1/3} / (V_o \tan \beta) \end{aligned} \quad (4.31)$$

Then, similarly to the result obtained by Carcaterra and Ciappi [36] for wedges,

$$F^*t^* = 5 \frac{2^{2/3} 7^{5/3}}{3^{14/3}} MV_o = 1.2067 MV_o \quad (4.32)$$

Experimental results were obtained for one circular plate and cone with small deadrise angles (less than 15 degrees). Simple formulas were obtained for the maximum pressure, and results are compared with those for two-dimensional wedges [133].

An approximation for small times can be obtained from Eq. (4.4), assuming that the added mass is that of a hemisphere of radius r

$$m = \frac{2\pi}{3} \rho (\zeta \tan \beta)^3 \quad (4.33)$$

Initially, $m \ll M$ and $\dot{\zeta} \approx V_o$, so $\zeta \approx V_o t$, and the hydrodynamic force can be approximated by

$$F = -M\ddot{\zeta} = \frac{M^3 V_o^2}{(M+m)^3} (2\pi \rho \zeta^2 \tan^3 \beta) \approx 2\pi \rho V_o^4 t^2 \tan^3 \beta \quad (4.34)$$

This result is identical to the 1951 theoretical prediction of Shiffman and Spencer [134]. Battistin and Iafrati [65] used a similar expression,

$$F \approx \frac{3\pi}{2} \rho V_o^4 t^2 \tan^3 \beta \quad (4.35)$$

Kleefman et al. [135] cited the formula of Shiffman and Spencer [134],

$$F = k(\beta) \rho V_o^4 t^2 \tan^3 \beta \quad (4.36)$$

which is also mentioned in an anonymous report [136].

4.2.2 Wagner's Approach for the Water Entry of Cones. In Wagner's theory, the pressure distribution under a plate of radius c moving with a velocity V is given by

$$\frac{p(x)}{\rho} = \dot{V} \sqrt{c^2 - x^2} + \frac{Vc}{\sqrt{c^2 - x^2}} \dot{c} - \frac{1}{2} \frac{V^2 x^2}{c^2 - x^2} \quad (4.37)$$

From the geometry of the cone during normal penetration of the water surface (Fig. 29), $c = \zeta \tan \alpha$, $\dot{c} = \dot{\zeta} \tan \alpha$, and $\dot{V} = \ddot{\zeta}$. After substitution, we find

$$\frac{p(x)}{\rho} = \ddot{\zeta} \sqrt{\zeta^2 \tan^2 \alpha - x^2} + \frac{\dot{\zeta}^2 \zeta \tan^2 \alpha}{\sqrt{\zeta^2 \tan^2 \alpha - x^2}} - \frac{1}{2} \frac{\dot{\zeta}^2 x^2}{\zeta^2 \tan^2 \alpha - x^2} \quad (4.38)$$

Defining $\bar{x} = x/\zeta$,

$$\frac{p(x)}{\rho} = \ddot{\zeta} \zeta \sqrt{\tan^2 \alpha - \bar{x}^2} + \frac{\dot{\zeta}^2 \tan^2 \alpha}{\sqrt{\tan^2 \alpha - \bar{x}^2}} - \frac{1}{2} \frac{\dot{\zeta}^2 \bar{x}^2}{\tan^2 \alpha - \bar{x}^2} \quad (4.39)$$

For a cone moving with a constant velocity ($\ddot{\zeta} = 0$),

$$\bar{p} = \frac{p(x)}{\rho \dot{\zeta}^2} = \frac{\tan^2 \alpha}{\sqrt{\tan^2 \alpha - \bar{x}^2}} - \frac{1}{2} \frac{\bar{x}^2}{\tan^2 \alpha - \bar{x}^2} \quad (4.40)$$

Poruchikov [137] analyzed the water entry of a cone using a similarity approach. De Backer et al. [138] used Wagner's approach (Eq. (2.23)) to calculate the pressure acting on the wetted surface of a cone or a hemisphere during water entry.

4.2.3 Oblique Impact of Cones. Gurney [139] considered the penetration of conical tipped missiles (Fig. 30). Following

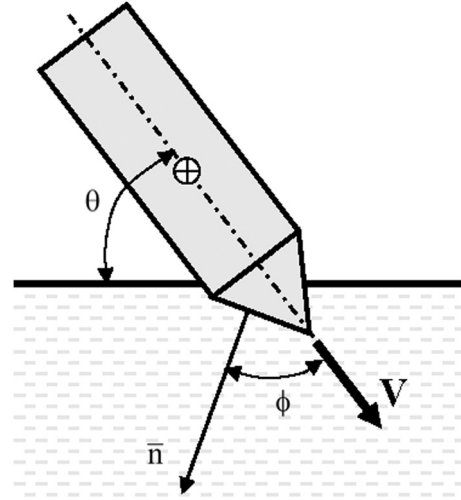


Fig. 30 Oblique impact of a projectile with a conical tip (Gurney [139])

Birkhoff, the pressure acting on an element dS of the wetted surface is assumed to be

$$p = C_1 \rho V^2 \cos^2 \phi \quad (4.41)$$

where ϕ is the angle between $\bar{n}$, the outside normal to dS , and the axis of the missile. This pressure acting on S_w , the wetted surface of the missile, is to create a force $\bar{F}$ and a moment $\bar{M}$ about the center of mass. Since it is not axisymmetric, $\bar{F}$ is not directed along the axis of the missile and $\bar{M}$ is not zero. This implies that the angle θ will change as the projectile rotates. This phenomenon is called whip. Park et al. [140] considered the ricochet behavior of a tangent ogive. Noonan and Steves [141] studied the trajectory of bullets fired into water and Mirshak [142] showed that the critical angle, above which ricochet does not occur, and the likelihood of stable or tumbling ricochets depend on whether waves are present and in what direction those waves are traveling relative to the path of the incoming bullet.

4.3 Water Entry of Cylinders. The following discusses the dynamics of the impact, the pressure distribution on the wetted surface of the cylinder, and the asymptotic expansion methods used to obtain accurate solutions for the early times of the event.

4.3.1 Von Karman's Approach for Cylinders. In the case of a horizontal cylinder, the added mass of the fluid per unit length is

$$m = \frac{\pi}{2} \rho r^2 = \frac{\pi}{2} \rho (2\zeta R - \zeta^2) \quad (4.42)$$

From Eq. (2.15), we find that the velocity of the cylinder is given by

$$\dot{\zeta} = V_o \left/ \left[1 + \frac{\rho}{\rho_s} \left(\zeta - \frac{\zeta^2}{2} \right) \right] \right. \quad (4.43)$$

where ρ is the density of the fluid, ρ_s is the density of the cylinder, and $\zeta = \zeta/R$ is the nondimensional displacement. After integration, Eq. (4.43) gives the following relationship between ζ and the nondimensional time $\tilde{t} = V_o t/R$:

$$\zeta + \frac{\rho}{\rho_s} \left(\frac{\zeta^2}{2} - \frac{\zeta^3}{6} \right) = \tilde{t} \quad (4.44)$$

Using the first part of Eq. (4.4), the acceleration is

$$\ddot{\zeta} = -\frac{(MV_o)^2}{(M+m)^3} \frac{\pi}{2} \rho (2R - 2\zeta) \quad (4.45)$$

where M is the mass of the cylinder per unit length. The hydrodynamic force (per unit length) can be written as

$$F = -M\ddot{\zeta} = \pi\rho V_o^2 R \left(1 - \tilde{\zeta}\right) / \left(1 + \frac{\rho}{\rho_s} \left(\tilde{\zeta} - \frac{\tilde{\zeta}^2}{2}\right)\right)^3 \quad (4.46)$$

The motion of the cylinder and the hydrodynamic force are defined by Eqs. (4.43)–(4.45). As the cylinder makes contact with the surface of the water ($\zeta = 0$), Eq. (4.45) shows that the initial value of the force is

$$F = \pi\rho V_o^2 R \quad (4.47)$$

and, as shown in Fig. 32, it is also the maximum hydrodynamic force applied to the cylinder during the impact. Initially, when $m \ll M$ and the velocity is constant ($\dot{\zeta} \approx V_o$), so $\zeta \approx V_o t$, the hydrodynamic force (per unit length) can be approximated by

$$F \approx \pi\rho V_o^2 (R - V_o t) = \pi\rho V_o^2 R (1 - \bar{t}) \quad (4.48)$$

This result for the initial stage of the entry of a circular cylinder is cited by several authors [135,143], but, as shown in Fig. 31, it is accurate only for very short times.

Cointe and Armand [25] discussed the fact that, in von Karman's approach, r is the radius at the intersection between the cylinder and the surface of the water, whereas Wagner [7] included a wetting factor of $\sqrt{2}$. That is, $r \rightarrow r\sqrt{2}$, and the force F in Eq. (4.44) will then be multiplied by two. Cointe and Armand showed that predictions of the initial force predicted by several authors fall between those of von Karman [6] and Wagner [7]. This is also reported in Molin's textbook [144]. Cointe and Armand [27] developed a matched asymptotic solution with an inner region near the intersection point and an outer region covering the rest of the fluid domain. The force predicted by the first order solution is that obtained using the conservation of momentum approach, including the wetting correction ($F \approx 2\pi\rho V_o^2 R$). A second order solution was also obtained, and it showed better agreement with experiments. Korobkin and Wu [145] developed a third order asymptotic solution for this problem.

The difficulty in obtaining accurate estimates of the force applied on a cylinder using numerical approach is discussed by Zhu et al. [146,147].

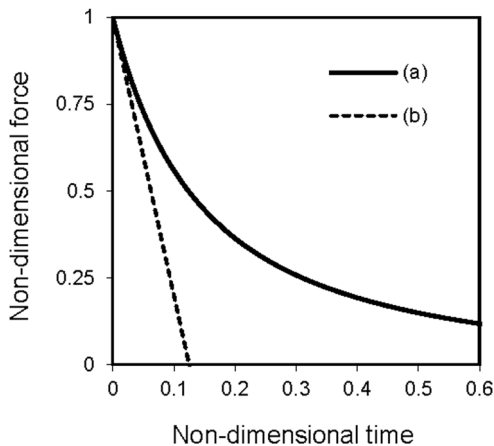


Fig. 31 History of the hydrodynamic force for a horizontal cylinder during vertical water entry in terms of the nondimensional force $F = F/(\pi\rho V_o^2 R)$ and the nondimensional time $\bar{t} = V_o t/R$ obtained from (a) Eqs. (4.44) and (4.46) and (b) Eq. (4.48)

4.3.2 Pressure Distribution During the Water Entry of a Cylinder. The pressure distribution on the wetted surface of a horizontal cylinder during water entry can be determined using Wagner's approach. Details will not be given here. Experimental results are available from several publications. Colicchio et al. [148] and Lin and Shieh [78,149] measured the pressure at several points on the circumference of a cylinder during a horizontal free fall into water. Lin and Shieh [78,149] noticed strong oscillations in the pressure measurements and believed that they may be caused by compression and expansion of the air layer trapped under the projectile during the early phase of the impact, based on the frequency of these oscillations. Colicchio et al. [148] also noted the effect of trapped air. Wardlaw [150] developed a numerical approach to predict impact pressures, forces, and moments during oblique water entry of axisymmetric solids.

4.4 Water Entry of Flat Plates. Many studies are concerned with the water entry of flat plates. Here, the phrase flat plate indicates that, initially, the surface of the plate is parallel to the free surface of the water. Recalling the previous discussion of wedges with very small deadrise angles or flat-bottom bodies, there can be some air-cushioning effects. Generally, this problem is avoided by considering small disks, for which such effects are negligible, or plates that are initially on the surface of the water and are suddenly set in motion, so there is no air trapped between the plate and the water. In the following, we will discuss how von Karman's approach can be applied to the impact of a plate on the water surface; we will discuss various asymptotic expansion approaches used for this problem, and the particular cases of impacts on shallow water and stone skipping will also be discussed.

Okada and Sumi [151] conducted experiments with plates impacting water with various deadrise angles. It was found that, for very low angles, the effect of trapped air is significant, the pressure is maximum in the center of the plate, and an empirical formula is given for predicting that maximum pressure. For larger angles, the behavior is predicted by Wagner's theory and the pressure is maximum near the intersection with the free surface, as we have seen in the discussion of the water entry of a wedge. There is also a transitional region, in which both patterns are observed. Note that Okada and Sumi [151] always tested two plates mounted as an asymmetrical wedge for nonzero deadrise angles. Faltinsen and Semenov [152] presented an analytical solution for a plate making an angle with the water surface and entering the water vertically or obliquely.

4.4.1 Von Karman's Approach for the Water Entry of Flat Plates. Glasheen and McMahon [153] conducted experiments on the vertical water entry of a small disk (radius: 31.7 mm, mass: 0.620 kg) and showed that the impact phase with a steep drop in velocity and a spike in the applied force is followed by a cavity running phase. Experiments confirmed that, for a circular plate, the added mass of the fluid is given by

$$m = \frac{4}{3} \rho r^3 \quad (4.49)$$

The duration of the impact phase is very short, and the sudden change in velocity can be predicted accurately using the conservation of momentum approach. A numerical analysis of these experiments was given by Gaudet [154].

For a plate of length $2L$ and unit width, the asymptotic analysis of Korobkin and Iafrati [155] showed that a simple analysis is possible. The velocity of the plate after the impact phase is determined using von Karman's momentum approach (Eq. (2.15)) with the following added mass:

$$m = \frac{\pi}{2} \rho L^2 \quad (4.50)$$

and the subsequent motion is governed by

$$(M + m)\ddot{\zeta} + 2.83\rho L^2 \zeta^2 \zeta^{-1/3} = 0 \quad (4.51)$$

In Korobkin and Iafrati's example, the plate is dropped from a height $h = 2$ m. The velocity before impact is $V_o = \sqrt{2gh} = 6.264$ m/s. The mass of the plate per unit width $M = 50$ kg and the length $L = 0.5$ m, so the added mass is 392.4 kg. With the momentum approach (Eq. (2.15)), the velocity after impact is $\dot{\zeta}(0) = 0.7075$ m/s. This prediction is in good agreement with the asymptotic approach. The velocity drops instantaneously from 6.262 m/s to 0.7075 m/s before any flow can be established. Often, rectangular plates are assumed to be very long in one direction, and a slice of unit width in that direction is considered. Payne [156] compared several expressions used for calculating the added mass of a rectangular flat plate of finite aspect ratio.

4.4.2 Asymptotic Expansion Methods for Analysis of Flat Plate Impact. Yakimov [157] was the first to study the jet flow close to the edge of a flat plate. In a series of articles, Iafrati and Korobkin [155,158–162] studied the two-dimensional problem of a plate that is originally floating on the free surface and then suddenly starts to penetrate the water with a constant entry velocity. In the approach presented by Iafrati and Korobkin [158], the flat plate impact problem is nondimensionalized so that the plate occupies the segment $-1 < x < 1$ on the water surface. The velocity of the plate is assumed to remain constant so that $y' = -Vt$, the actual vertical position of the plate, varies according to $y' = -Vt$. In the nondimensional formulation, the velocity of the plate is -1 and the position is given by $y = -t$. For incompressible and irrotational fluid flow, the velocity potential should satisfy the differential equation

$$\Phi_{xx} + \Phi_{yy} = 0 \quad (4.52)$$

with the boundary condition

$$\Phi_y = -1 \quad (|x| < 1, y = -t) \quad (4.53)$$

which states that the velocity of the plate is -1 in the y -direction. On the free surface of water,

$$\zeta_t + \Phi_x \zeta_x = \Phi_y \quad (4.54)$$

$$2\Phi_t + \Phi_x^2 + \Phi_y^2 = -\frac{2}{Fr^2} \zeta + \frac{2}{We^2} \frac{\zeta_{xx}}{(1 + \zeta_x^2)^{3/2}} \quad (|x| > 1, y = \zeta(x, t)) \quad (4.55)$$

where ζ is the elevation of the free surface relative to its initial level, which should satisfy the boundary conditions

$$\zeta(\pm 1, t) = -t, \quad \zeta_x(\pm 1, t) = 0 \quad (4.56)$$

on the edges of the plate and the initial conditions

$$\zeta(x, 0) = 0, \quad \zeta_x(x, 0) = 0 \quad (4.57)$$

$$\Phi \rightarrow \infty, \quad \text{when} \quad x^2 + y^2 \rightarrow \infty \quad (4.58)$$

Outside a small region near the edge of the plate, the solution is sought in terms of the small time asymptotic expansion for the velocity potential and the height of the free surface

$$\Phi(x, y, t) = \Phi_0(x, y) + t\Phi_1(x, y) + t^2\Phi_2(x, y) + \dots \quad (4.59)$$

$$\zeta(x, y, t) = t\zeta_0(x, y) + t^2\zeta_1(x, y) + t^3\zeta_2(x, y) + \dots \quad (4.60)$$

The first order solution of the outer problem gives

$$\Phi_0(x, 0) = -\sqrt{1 - x^2} \quad \text{for} \quad |x| < 1 \quad (4.61)$$

$$\zeta_0(x, 0) = \frac{|x|}{\sqrt{1 - x^2}} - 1 \quad \text{for} \quad |x| > 1 \quad (4.62)$$

Near the edge, the “inner solution” is obtained using the stretched variables $x = 1 + Bt^{2/3}$, $y = Bvt^{2/3}$, and $B = (9/2)^{1/3}$. The velocity potential and the height of the water surface are taken as

$$\Phi = \sqrt{2B}t^{1/3}\varphi(u, v, t) \quad \zeta = Bt^{2/3}\eta(u, t) \quad (4.63)$$

Substituting Eq. (4.63) into Eqs. (4.52)–(4.57) leads to a boundary value problem for the inner problem. Solving this problem and matching the inner and outer solutions give a first order asymptotic expansion solution. A first order asymptotic solution for the water impact of a circular plate was developed by Bulgarelli et al. [159].

Iafrati and Korobkin [160,161] developed a second order for the same problem with the added complication that h , the nondimensional penetration of the plate, is now an arbitrary function of time, so that, under the plate, Eq. (4.53) is replaced by

$$\Phi_y = -h_t \quad (|x| < 1, y = -h) \quad (4.64)$$

A solution is sought using the second order expansion

$$\Phi(x, y, t) = h_t(t)\Phi_0(x, y) + h(t)h_t(t)\Phi_1(x, y) + o(h) \quad (4.65)$$

$$\zeta(x, t) = h(t)\zeta_0(x, y) + h^2(t)\zeta_1(x, y) + o(h^2) \quad (4.66)$$

If $h = -t$, Eqs. (4.65) and (4.66) reduce to Eqs. (4.59) and (4.60). These authors used the same approach to obtain a second order solution for a circular disk [162]. A similar approach was used by King et al. [163,164]. Korobkin [165] studied a related problem of a plate floating on a liquid subjected to a given external load using an asymptotic approach. This study is motivated by the problem of a spacecraft being launched from an artificial floating island.

4.4.3 Impact of Flat Plates on Shallow Water. Korobkin [98] considered the impact of a boxlike structure on shallow water and examined the acoustic phase of the impact (Fig. 16) and the cavity forming phase (Fig. 17). Tkacheva [166] considered a box with an elastic bottom impacting a thin liquid layer. Korobkin and Ohkusu [167] studied the impact of two circular plates, one of which is floating on a thin layer of liquid. This problem was also investigated by Ermanyuk and Ohkusu [168]. In the experiments of Ermanyuk et al. [169] disks with flat, convex, or concave bottoms impact shallow water.

Ryabchenko [170] describes a method of integral equations for solution of the 2D and 3D problems of plate impact on an incompressible fluid of finite depth. Solutions of the equations are obtained analytically and numerically using the collocation method. Initially, the plate is floating on the surface of the water and, at time $t = 0$, an impact gives it a velocity V_o . The subsequent motion of the rigid plate is determined using this approach.

4.4.4 Skipping Stones. A related problem that has received some attention is the skipping of stones on the surface of the water. This problem has been studied by considering a spinning circular disk that makes an angle ϕ with the water surface (Fig. 32). The velocity of the center of mass makes an arbitrary angle θ with the water surface. Bocquet and Clanet [171] and Clanet et al. [172] presented experimental results and Bocquet [173] and Rosellini et al. [174] analytical models. Nagahiro and Hayakawa [175] studied this problem using the SPH method, and another research group [176–178] analyzed the motion of nonspinning plates impacting the water with an arbitrary angle of attack. Park et al. [140] used a numerical approach based on the source panel method to study the ricochet of a circular disk and that of an ogive-shaped body.

4.5 Water Entry of Bodies With Various Other Shapes.

While the literature focused mostly on the simple geometrical shapes previously discussed, other shapes have also been

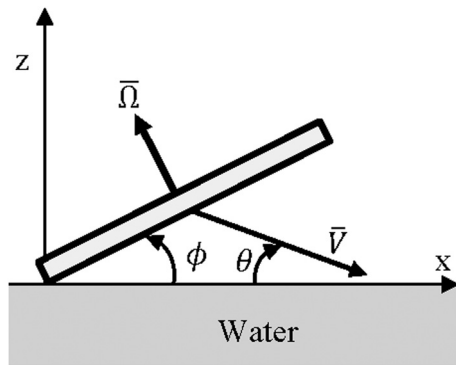


Fig. 32 Model for stone skipping problem (adapted from Nagahiro and Hayakawa [175])

considered: ellipsoid [179,180], pyramid [181], elliptic paraboloid [182–185], and flared bow area [186–189]. Zhao and Faltinsen [190] discussed asymmetric water entry of a bow-flare section.

5 Water Entry of Elastic Structures

During the water entry of an elastic structure, deflections induced by the pressures generated on the wetted surface could significantly affect the problem. Considering a panel on the hull of a ship during hull slamming, some insight can be gained by considering a one mode approximation for calculating its response to a half sine pulse of period T_L and amplitude F . The modal equation can be written as

$$m\ddot{x} + kx = F \sin \frac{2\pi t}{T_L} \quad (5.1)$$

where m and k are the effective mass and the stiffness. $T_N = 2\pi\sqrt{k/m}$ is the natural period of the system. Defining the nondimensional displacement $\bar{x} = kx/F$ and the nondimensional time $\tau = 2\pi t/T_L$, the response can be written in terms of a single parameter T_L/T_N as

$$\bar{x} = \left[\sin(\tau) - \frac{T_N}{T_L} \sin \frac{T_L}{T_N} \tau \right] / \left[1 - \left(\frac{T_N}{T_L} \right)^2 \right], \quad \text{when } \tau \leq \pi \quad (5.2)$$

When the period of the loading is much larger than the natural period ($T_L \gg T_N$), Eq. (5.2) shows that the response tends to $\bar{x} = \sin \tau$, the quasistatic response. When $T_L \ll T_N$, Fig. 33 shows that, during the impact phase, the deflections are very small. Therefore, if a rigid body analysis predicts that the loading phase lasts less than one-tenth of the period of the vibration mode excited by the pressure loading ($T_L < T_N/10$), an uncoupled analysis is justified. In other words, a first analysis can be performed to determine the pressure distributions, assuming that the body is rigid. These pressures are then used in a separate analysis to calculate the dynamic response of the panel. If $T_L > T_N/10$, a coupled analysis might be required to obtain an accurate prediction of the pressures acting on the panel.

5.1 Water Impact of Flat Plates. Korobkin and Khabakhpasheva [191] studied the problem of a plate initially in contact with water. The plate is assumed to be the bottom of a structure and is modeled as a Bernoulli–Euler beam, and the displacements are written using the modal expansion method. The plate deflections are determined using Wagner’s theory. Several investigators considered flexible wedges, in which the sides were sandwich plates with foam cores and composite facings. Charca et al. [192–194] studied the damage induced by repeated water impacts and Das

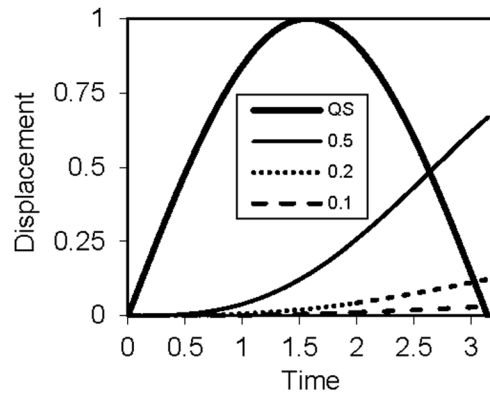


Fig. 33 Transient response of a SDOF system to a sine pulse: nondimensional displacement versus nondimensional time (Eq. (5.2)). Quasistatic response QS and response for three values of the ratio T_L/T_N : 0.5, 0.2, 0.1.

and Batra [195] developed a numerical approach to predict the stresses in the sandwich plates during the impact.

5.2 Water Impact of Shells

5.2.1 Water Impact of Cylindrical Shells. Sun and Faltinsen [189] used the boundary element method (BEM), cylindrical shell modeled as a ring. The water impact of cylindrical shells was studied by Arai and Miyauchi [196,197] using a coupled analysis, in which a modal analysis approach was used for the shell and a computational fluid dynamics approach for the fluid domain. Ionina [198] and Ionina and Korobkin [199] considered the impact of a horizontal isotropic cylindrical shell on water. The deflections of the shell were calculated using a modal superposition technique and were determined by solving the coupled fluid flow-shell deflection problem. The shell is assumed to be inextensible in the circumferential direction, and the duration of the impact is expected to be much shorter than the period of the second mode of vibration of the shell. Sun [200] and Sun and Faltinsen [189] studied the water impact of an elastic cylindrical shell as a coupled problem where the fluid flow and the deformation of the shell are solved simultaneously. The boundary element method is used to model the fluid flow, and a modal expansion approach is used for calculating the structural response. Khabakhpasheva [201,202] studied the impact of a cylindrical shell on a thin layer of water. The modal superposition approach is used for modeling the shell, and the matched asymptotic expansion approach of Korobkin [203] is used for the fluid domain. In that expansion, the fluid is divided into four regions (Fig. 34).

5.2.2 Water Impact of Spherical Shells. As previously mentioned, Cappelli et al. [114,116] studied the water landing of Apollo capsules, the bottom of which are spherical shell panels. In the first report, the capsule is treated as a rigid body to determine the pressure acting on the wetted area, and then those pressures

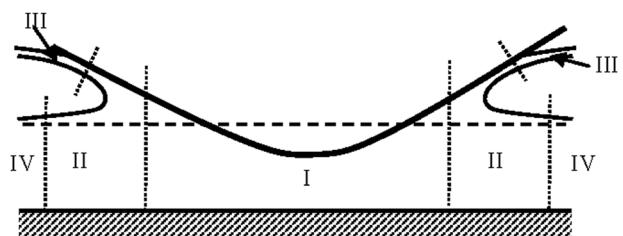


Fig. 34 Shallow water entry of a rigid body (adapted from [201,202]). I: region beneath the entering body; II: the jet root, III: the spray jet; IV: the outer region.

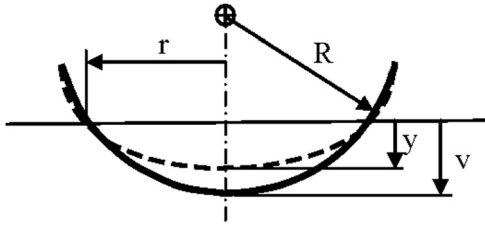


Fig. 35 Elastic spherical shell impacting the water surface (adapted from Bingman [210]). Solid line: initial shape; dashed line: deformed shape.

are used to calculate the dynamic response of the shell. In the second report, a coupled analysis is presented. The same general problem was studied by Gorshkov and Grigolyuk [204]. Scolan [205,206] dealt with the free-fall water entry of elastic cones and, as in the previous three references, the shell coming in contact with the water is connected to a much heavier rigid body. Results show that the elasticity of the shell affects the velocity and acceleration of the structure as the penetration depth becomes significant. Constantinescu et al. [207,208] also studied the water entry of elastic cones and used the finite element method for both the fluid and the solid domains.

Payton [209] examined the stresses in cylindrical and spherical elastic shells impacting a fluid. The shells are approximated by beam and flat plate models subjected to expanding pressure loads. Bingman [210] introduced a modified momentum approach to account for the deformation during the normal impact of a spherical shell on the free surface of water. As shown in Fig. 35, δ_0 is the maximum deflection of the immersed portion of the shell. As in our previous notation, V_0 is the initial velocity of the sphere, M is its total mass, and m is the added mass of the fluid. Bingman assumed that the total mass can be divided into two parts: M_R , the mass of the upper part of the shell that remains essentially undeformed and M_S , the mass of the wetted part of the shell. M_R is assumed to be concentrated at the center of mass of the undeformed shell, and M_S is assumed to be concentrated at the lowest point of the shell. With this notation, conservation of linear momentum is given by

$$MV_0 = M_R \dot{y}_1 + M_S (\dot{y}_1 - \dot{\delta}_0) + m (\dot{y}_1 - \chi \dot{\delta}_0) \quad (5.3)$$

where χ is a number between 0 and 1.0, which reduces the center deflection velocity to the mean or effective velocity of the wetted surface. It was shown that, for spherical shells, χ is approximately equal to $\frac{2}{3}$. The deflection was calculated using a formula for the static deflection of the wetted portion of the shell. The flexibility of the shell resulted in oscillations in the velocity of its center of mass. The predictions from that model were shown to be in good agreement with experimental results. A similar two degree of freedom model was used by Gorshkov and Grigolyuk [204].

6 Numerical Methods

This review emphasized the use of analytical methods for studying the water entry problem as a way to emphasize the physics of the problem and the difficulties encountered in its analysis. However, many authors use numerical models, and this section provides references to numerical analyses of the water entry problem. Most numerical models assume that the water is inviscid, irrotational, and incompressible, and the flow is governed by Laplace's equation. The effects of gravity and surface tension on fluid motion are also neglected in most studies. The validity of these assumptions is assessed in some of the references.

This section lists 66 references dealing with numerical approaches. Approaches based on boundary elements, smooth par-

ticle hydrodynamics, the finite element method, and fixed grid methods are described in some detail in Secs. 6.1–6.4.

6.1 Boundary Elements. Several authors used the boundary element method to model the fluid domain during the water entry of rigid bodies. Zhao and Faltinsen [190] presented a nonlinear BEM for two-dimensional bodies with an arbitrary cross section with a jet flow approximation and applied this approach to rigid wedges with deadrise angles between 4 deg and 81 deg. Sun and Faltinsen [189] used this approach to study the water impact of horizontal circular cylinders and cylindrical shells. Zhao and Faltinsen [190] considered the water entry of a bow flare ship section with a constant roll angle. Battistin and Iafrati [63] used the BEM to describe the flow and pressure distribution outside the thin jet layer for wedge impacts. In that work, the entry velocity is kept constant during the impact. In another article [65], these authors apply the same approach to the water entry of cones, circular cylinders, and spheres. Xu et al. [211] is said to be the first numerical analysis of the water entry of a cone in free fall. Xu et al. [87] use the BEM for the water entry of asymmetrical wedges. Most studies consider impacts on deep water, but Lin and Ho [212] used the BEM for the 2D entry of a wedge in shallow water depths.

The effect of gravity on fluid flow is assessed by considering the nondimensional Froude number $Fr = \sqrt{V/(gt)}$, where V is the velocity, g is the gravitational acceleration, and t is the time. Most analyses assume that gravity effects are negligible as the Froude number becomes very large, that is, when V/g is much larger than the duration of the impact. A study of cones and spheres striking the surface of the water at a constant velocity indicated that the shape of the jet is strongly affected by gravity when Fr is near one [211–215]. Kihara [216] also indicated that gravity effects are important in the jet region.

Lu et al. [217] presented a coupled hydroelastic analysis of the water impact of flexible wedges in which the water is modeled using the BEM and the structure is modeled using beam finite elements. The jet is cut off and modeled following the approach in Zhao and Faltinsen [190]. Small oscillations of the sides of the wedge are predicted, but there does not appear to be any coupling between the impact and the oscillation of the side panels for the examples considered.

6.2 SPH Method. Smoothed particle hydrodynamics (SPH) is a method introduced by Gingold and Monaghan [218] in 1977 to obtain approximate numerical solutions of the governing equations by replacing the continuum under study with a set of particles. This method has been applied to many fields and Monaghan has published extensive reviews of its use [219–221]. SPH is the oldest in a class of methods called meshfree particle methods (MPMs) or meshfree methods. An introduction to the SPH method is given by Liu and Liu [222], and a more general view of meshfree methods can be found in Li and Liu [223] and Liu [224].

The application of SPH to the water entry of rigid and elastic bodies is reviewed next. Many of the approaches discussed can be applied to the analysis of wet deck slamming, wave slamming, or the sloshing of liquids in a tank. However, these topics are beyond the scope of this review and will not be discussed in this section.

6.2.1 SPH Simulations of Water Entry of Rigid Bodies. The SPH method has been used to study the water entry of rigid wedges [225–236], sections of flared monohull structures [225], cylinders (e.g., Ref. [226]), and other shapes. Often, rigid wedge problems are treated in order to show that SPH and its implementation in a particular software package can provide accurate solutions by comparison with Wagner's solution. SPH is able to model complicated free surface flows, including jets and splashing during the water entry of a solid object [225]. Numerical analyses with SPH can handle various geometries and provide detailed results for the complex flows at the intersection between the body and the free surface of the water.

Two issues arise with SPH: (1) the size of the size step and (2) reflection of waves from the boundaries of the computational domain. For explicit time stepping, the time step is normally controlled by the Courant, Friedrichs, Lewy (CFL) condition. For SPH simulations, this condition states that the time step is limited by the values of the smoothing length h and c and the speed of sound in the fluid

$$\Delta t = kh/c \quad (6.1)$$

where k is a constant. The smoothing length is related to the spacing of the particles and should be small enough to capture details in areas with high gradients. In Ref. [233], the water impact of a wedge with a base length of 500 mm is analyzed using 4.5×10^6 particles with an initial spacing of 1.2 mm.

The stress tensor can be expressed as the sum of a hydrostatic stress tensor and a deviatoric stress tensor. For a fluid, $\sigma_{ij} = -p\delta_{ij} + s_{ij}$, where p is the pressure, δ_{ij} is the Kronecker delta function, and s_{ij} and s_{ij} are components of the stress tensor and the deviatoric stress tensor, respectively. For Newtonian fluids, the deviatoric stress tensor is directly proportional to the dynamic viscosity μ and

$$\sigma_{ij} = -p\delta_{ij} + \mu(1 - \delta_{ij}/3)\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) \quad (6.2)$$

where $\bar{v}$ is the velocity vector. The first term accounts for bulk compression and the second term for shear. In the analysis of water impact problems, the effects of viscosity are usually negligible, and in the remainder of this section, $\mu = 0$, unless stated otherwise. The pressure p is related to changes in the density of the fluid using an equation of state (EOS). For the problem under consideration here, SPH analyses most often use Tait's equation of state [225,227–229,231,233,234,237]

$$p = B[(\rho/\rho_0)^n - 1] \quad (6.3)$$

where $B = \rho_0 c^2/n$, ρ_0 is the initial density of the fluid, ρ is the current density, and n is a constant. For sea water, $n = 7$, $\rho_0 = 1000 \text{ kg/m}^3$, and $c = 1500 \text{ m/s}$. For air, $n = 1.4$, $\rho_0 = 1.29 \text{ kg/m}^3$, and $c = 343 \text{ m/s}$. This equation of state is appropriate for weakly compressible air flows. Analytical solutions are derived assuming that the water is incompressible. Other equations of state can be used as well. For example, Panciroli et al. [109] used the Gruneisen EOS. Several studies (e.g., Ref. [231]) have shown that, when the Mach number (particle velocity divided by speed of sound) is less than 0.1, the fluid can be considered to be incompressible. In order to increase the size of the time step allowed by the CFL condition, several authors use a much lower speed of sound in their simulations, as suggested by Monaghan [238]. For water, Viviani et al. [227] use a speed of sound in the 50–250 m/s range, while Oger et al. [231–233] use 80 m/s and Veen and Gourlay [235] use $c = 100 \text{ m/s}$. Oger et al. [231] use $c = 150 \text{ m/s}$ instead of 340 m/s for air. This is done by changing the value of B in Eq. (6.2) while using the same initial density and the same n . A lower value of c then results in a higher time step (Eq. (6.1)). This is acceptable because, for low velocity impact problems, wave effects are not important, and the analysis focuses on the flow of water surrounding the solid. Some authors (e.g., Ref. [231]) also used artificial viscosity terms suggested by Monaghan [238] in their formulation in order to increase step size while maintaining stability.

The size of the fluid region included in the model is limited in order to keep computational costs down. Then, acoustic waves generated by the impact of the solid as it enters the water will reflect from the boundaries of the computational domain and propagate back towards the point of impact and perturb the solution. Several approaches are used to provide nonreflecting boundaries [228].

Oger et al. [231,232] and Gong et al. [229] studied impacts in which the wedge becomes completely submerged and is followed

by an enclosed air cavity. Both the water and the air were modeled with SPH. In Ref. [232], the air above the water surface is also included in the model in order to capture air cushioning effects before the wedge enters the water.

References [239–241] studied a billiard ball entering the free surface of the water with an initial vertical velocity of 4.8 m/s. SPH predictions were shown to be in good agreement with experimental results, but this study illustrates a significant issue with the method: a very large number of particles is needed. For this simple problem, the simulations described in Ref. [239] had between 51,717 and 10,161,291 particles. The article mentions other simulations with up to 55×10^6 particles. For comparison purposes, in the simulation of the water entry of rigid wedges, Ref. [231] uses 230,000 air and water particles, 2×10^6 particles are used in Ref. [228], and 4.5×10^6 s in Ref. [229]. Bøckmann et al. [242] studied the water entry of a rigid rectangular box and discussed the effect of the number of particles on the accuracy of the solution. Several articles mention the large computational costs associated with SPH (e.g., Ref. [228]). Both Refs. [239] and [240] discuss how the SPH method can be implemented on parallel processors in order to speed up calculations.

6.2.2 SPH Simulations of Water Entry of Flexible Structures.

For practical applications, there is interest in modeling the motion of the fluid and the deformation of the structure in order to predict any possible interaction. Usually, when SPH is used to model the fluid, the finite element method (FEM) is used to model the structure. In recent publications, both solids and fluids are discretized using SPH.

SPH was used to study hydroelastic effects during the water entry of elastic beams. Fourey et al. [243] conducted experiments on the water entry of elastic wedges with a deadrise angle of 10 deg. The wedge is an isosceles triangle, whose base is 1.2 m long. Each side of the wedge is a 4-mm-thick, aluminum, cantilever beam. The wedge enters the water at 30 m/s. Pressure sensors measure the pressure history at four locations along one side of the wedge, and SPH simulations appear to be in good agreement, except for the peak pressure. Panciroli et al. [109] conducted similar experiments for free-falling elastic wedges impacting the free surface of water. The dimensions of the side panels are selected in order to generate a definite fluid-structure interaction. The period for the first vibration mode of the side panels was selected to be of the same order as the impact duration determined with von Karman's approach for rigid wedges. The analysis shows that a series of separations and impacts between the panels and the fluid strongly affects the dynamics of the impacting body.

Groenenboom et al. [244,245] used a coupled SPH-finite element (FE) approach to predict the fluid-structure interaction (FSI) effects during the impact of a flexible cylinder on calm water. The SPH method was used to model the fluid, and the finite element method was used for the structure. A coupled SPH-FE was also used by Grimaldi et al. [246] to study the ditching of a transport plane.

Several authors study fluid structure interaction problems by modeling both the fluid and the structure using SPH (e.g., Refs. [247,248]). Oger et al. [249,250] used this approach to study the impact of beams on the surface of the water.

6.3 Finite Element Method. Several studies use the finite element method to analyze the impact of rigid or flexible bodies on the free water. Ribet [251] used a finite element model to study the water impact of spheres, flexible wedges, and ellipsoidal bodies. Donguy et al. [183] used the FEM to model the fluid domain, assuming potential flow, and determined the response to an impact on the free surface by rigid bodies: a wedge, a cone, and an elliptic paraboloid. In the case of deformable bodies, the FEM was also used to model the solid. Peseux et al. [184] used the finite element method to analyze the water entry of a parabolic hull, a rigid cone, and an elliptic paraboloid. Stenius et al. [252–256] performed finite element simulations of the water entry of wedges

using the commercial software package LS-DYNA and concluded that it is computationally very expensive to resolve the absolute pressure peak near the intersection point. However, they added that, when studying the structural response and the fluid-structure interaction, the value of this pressure peak is of less importance [256]. Reference [252] shows that hydroelastic effects depend on the ratio $R = \text{TLP}/\text{TNP}$, where TLP is the loading period and TNP is the wet natural period. For a wedge with side panels of length b ,

$$R = 4 \left(\frac{\mu_{\text{NP}}}{\pi} \right)^2 \frac{\tan \beta}{V} \sqrt{\frac{D}{\pi \rho_w b^3}} \quad (6.4)$$

where μ_{NP} is a constant, depending on the boundary conditions; V_0 is the initial velocity, β is the deadrise angle, ρ_w is the density of water, and D is the bending rigidity of the panel. It was found that there are no hydroelastic effects when $R > 4$.

Lewis et al. [257] used ANSYS full three-dimensional Reynolds-averaged Navier–Stokes equations (RANSE) to study the water entry of a wedge. Aquelet et al. [258–260] used a finite element approach, in which the structure is modeled using a Lagrangian approach and the fluid using an Eulerian or arbitrary Lagrange–Euler (ALE) approach, and a coupling algorithm is introduced at the interface to avoid oscillations in the results. The water entry of a wedge is considered as an example.

Das and Batra [195] used LS-DYNA to model the water impact of a wedge consisting of two sandwich panels joined at an angle. The 2D analysis is performed using rectangular elements for the solid domain and the ALE approach developed by Aquelet et al. [258–260], which is implemented in that software package.

6.4 Fixed Grid Methods. In fixed grid methods, such as finite difference and finite volume methods, where the water, the air, and the impacting solid are interacting, special techniques are needed to detect the interface between the different domains. Three such methods are: the cubic interpolated pseudoparticle (CIP) method, the volume of fluid (VOF) method, and the level set method. The application of these methods to the analysis of water entry problems is discussed next.

6.4.1 CIP Method. The cubic interpolated pseudoparticle (CIP) method was used by several investigators [147,261–267] to capture the free surface of the water during impact. The three-dimensional motion of a compressible viscous fluid is governed by the conservation of mass equation,

$$\frac{\partial \rho}{\partial t} + u_i \frac{\partial \rho}{\partial x_i} = -\rho \frac{\partial u_i}{\partial x_i} \quad (6.5)$$

the conservation of momentum equation,

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\rho \frac{\partial \sigma_{ij}}{\partial x_j} + f_i \quad (6.6)$$

the constitutive equation (Eq. (6.2)), and an equation of state of the form $p = f(\rho)$, such as Eq. (6.3). Using the EOS, the conservation of mass equation (Eq. (6.5)) becomes

$$\frac{\partial p}{\partial t} + u_i \frac{\partial p}{\partial x_i} = -\rho c^2 \frac{\partial u_i}{\partial x_i} \quad (6.7)$$

where the speed of sound is given by $c = \sqrt{\partial p / \partial \rho}$. The problem is discretized using the finite difference method solved using a fractional step approach consisting of three phases: an advection phase, in which the right-hand side of Eqs. (6.5)–(6.7) is set to zero, and two nonadvection phases. In the advection phase, approximate values of the spatial derivatives of the density, velocity, and pressure are interpolated by a polynomial inside upwind cubic cells defined by the computational grid. This explains the terms

“cubic interpolated” in CIP. Details about the method can be found in the references cited.

Zhu et al. [147] studied the water entry of cylinders. Yang and Qiu [261] studied the water entry of 3D wedges. In most studies, the impacting wedge is a 2D solid with a triangular cross section, which implies that the solid is infinite in the third direction. In Ref. [261], the wedge has a finite length in the third direction. Reference [262] uses the CIP method to study the water entry of 2D wedges with both large and small deadrise angles, the water entry of a catamaran, and the water entry of finite cylinders (3D analysis). The water entry of spheres and 2D wedges were also studied using the CIP method in Ref. [263]. For small deadrise angles, the compressibility of air between the bottom of the wedge and the free surface was modeled. Tajima and Yabe [267] used the CIP method to investigate the effect of air cushioning during water entry of bodies with small deadrise angles. Other examples are provided in Refs. [264–266].

6.4.2 Volume of Fluid Methods. Scardovelli and Zaleski [268] reviewed the literature dealing with the VOF method and the level set method in the study of interfacial flows and free surface flows. The VOF is another technique used to determine the location of interfaces. It was introduced by Hirt and Nichols [269] in 1981 and used by several authors [135,143,270,271] to study water entry problems. Kleefsman et al. [135] used the VOF method for wedges, cylinders, and cones entering the water at prescribed velocities. Mnasri et al. [270] and Panahi et al. [271] used the method for the impact of horizontal cylinders. The Navier–Stokes equations for the motion of the fluid are discretized using the finite volume method, and the VOF method is used to locate interfaces inside each finite volume.

6.4.3 Level-Set Methods. A level set method is a numerical technique for tracking interfaces that is used rather extensively in fluid mechanics. For example, Sussman et al. [272] used this method to track the surface of a bubble in a liquid. The interface is a surface defined by a function $\phi(\bar{x}) = 0$, where $\bar{x}$ defines the position of a point in space. The evolution of that surface is governed by the equation

$$\frac{\partial \phi}{\partial t} + (\bar{v} \cdot \bar{\nabla}) \phi = 0 \quad (6.8)$$

For incompressible viscous fluids, the density ρ and the viscosity μ follow Eq. (6.8) like the function ϕ [272].

Finite difference discretization of fluid domains and the level set methods were used to study water impact problems [187,273,274]. A level-set method was used by Yang and Stern [273] to study the water entry and exit of horizontal cylinders and the water entry of a wedge sliding on a slope. Zhang et al. [274] also used this approach for the water entry of cylinders.

7 Related Problems

Two related water entry problems are the launching of lifeboats and the motion of mines impacting the surface of the water and falling to the bottom of the sea.

7.1 Lifeboats. Lifeboats can either be lowered into the water by suspension lines or skid on a ramp and free fall into the water. The task is to study the dynamics of the boat and then the effects of the acceleration on the passengers. In modeling the water entry of lifeboats lowered by suspension lines from a boom, Raman-Nair et al. [275] consider the effects of gravity, buoyancy, and suspension line tensions but make no attempt to model the impact forces at splashdown. Reaz et al. [276] compare two modes of launching lifeboats: from a skid or from a hook. Reaz and Khondoker [277] and others [278] model the launching of lifeboats by the free fall method in four phases: sliding on the skids, rotation at the end of the skids, free fall, and water entry, following the work

of Boef [279,280]. During the rotation phase at the end of the skids, the forces acting on the lifeboat are the weight, the normal force and the frictional force at the point of contact with the skid, and the lift and drag forces due to the wind. The wind can also induce a moment on the lifeboat during both the rotation phase and the freefall phase. The acting of the wind can have a significant effect on the attitude of the boat as it impacts the water surface. Willis et al. [281] focused on the effect of wind on the free fall phase, including the effects of lift, drag, and wind-induced moments. The location of the center of gravity of the lifeboat affects its angular velocity during the free fall phase, the angle of entry into the water, and the accelerations [282].

Different organizations, such as the International Maritime Organization (IMO), require that the injury potential of the acceleration be investigated. Nelson et al. [283] measured the accelerations of a lifeboat launched using the free fall method. Requirements of the Safety of Life at Sea (SOLAS) of the International Maritime Organization (IMO) are that the allowable accelerations in the x, y, and z directions should not exceed 18 g, 7 g, and 7 g for conventionally launched lifeboats. For free-falling lifeboats, the corresponding allowable accelerations are 15 g, 7 g, and 7 g, respectively. The xyz coordinate system is associated with the seats in the lifeboat. The x-axis is perpendicular to the chest and is positive towards the front, the y-axis is parallel to the shoulders and positive towards the left, and the z-axis is parallel to the spine and positive upward.

Nelson et al. [284] use these criteria to evaluate the potential for injury to the occupants. Nelson et al. [285] discuss several criteria used to predict injuries to humans, including the square-root-sum of the squares acceleration criterion,

$$\sqrt{\left(\frac{A_x}{15}\right)^2 + \left(\frac{A_y}{7}\right)^2 + \left(\frac{A_z}{7}\right)^2} \leq 1 \quad (7.1)$$

where A_x , A_y , and A_z are the linear accelerations in the x, y, and z directions expressed in g. This criterion defines a safe domain with an ellipsoidal envelope. It is based only on the amplitude of the acceleration and does not account for the duration of the acceleration pulse. Head injury criteria are also used. For example, Gadd's severity index,

$$SI = \int_0^T (a(t))^2 dt \quad (7.2)$$

where $a(t)$ is the resultant acceleration (in g) at the center of gravity of the head and T is the duration of the acceleration pulse (in seconds). This index accounts for the duration of the pulse, and a threshold value of 1000 is used to predict concussions in frontal impacts. The head injury criterion (HIC) used by the National Highway Transport Safety Administration

$$HIC = \left[(t_2 - t_1) \left\{ \frac{1}{(t_2 - t_1)} \int_{t_1}^{t_2} a(t) dt \right\}^{2.5} \right]_{\max} \quad (7.3)$$

With this criterion, a sort of moving average, where the moving time window $t_2 - t_1$ (in seconds) is usually taken as 0.036 s, is evaluated, and its maximum value should not exceed 1000 to avoid concussions. Nelson et al. [285] also used the multiaxial dynamic response method criteria, which is also known as the Brinkley dynamic response model and was developed during the Apollo space program [286,287] and is still used today for the design of the next generation manned space capsule named Orion [288–291]. In this approach [288–290], a lumped parameter model of the human body is used, and the motion is governed by three second-order differential equations of the form

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2x = A(t) \quad (7.4)$$

where x is the motion relative to the seat, ω_n is the undamped natural frequency, ξ is the damping ratio, and A is the measured acceleration of the seat. Nelson et al. [285] and Lawrence et al. [288] used the following values: $\omega_x = 62.8$, $\omega_y = 58.0$, $\omega_z = 52.9$, $\xi_x = 0.2$, $\xi_y = 0.09$, and $\xi_z = 0.224$. In each direction, the dynamic response is defined as

$$DR = \omega_n^2 x / g \quad (7.5)$$

which is the normalized response to the input acceleration in g. The maximum value of $DR(t)$ for any axis should be less than the limiting value. Limiting values for low, moderate, and high risk of injuries are given in Table 2 of Nelson et al. [285] for all three directions. In general, the risk of injury is different in each direction, and that it is different in the $+z$ and the $-z$ directions. Since accelerations usually occur simultaneously in all three directions, the following elliptical criterion is used:

$$\sqrt{\left(\frac{DR_x(t)}{DR_x^{\lim}}\right)^2 + \left(\frac{DR_y(t)}{DR_y^{\lim}}\right)^2 + \left(\frac{DR_z(t)}{DR_z^{\lim}}\right)^2} < 1 \quad (7.6)$$

Tabiei et al. [291] also used many other criteria that are commonly used in the analysis of automotive crashes to predict specific types of injuries: the neck injury criterion, the chest criterion, and the femur force criterion, in addition to the HIC previously discussed.

7.2 Mines. Another related problem is that of the free fall of large cylinders in air, water, and sediment. It is motivated by the need to predict the motion of mines that are dropped from airplanes or ships and get buried in sea floor. The 3D dynamic behavior of the rigid cylinder is described by the three components of the linear velocity of the center of mass and three angular velocities. An important goal is to predict the amount of burial of the cylinder in the ocean floor. A recent review of the literature was presented by Chu [292].

In the Kirchhoff–Kelvin theory developed in the 19th century, the body is subjected to its own weight W , a buoyancy force B , and drag force F_D ,

$$W = \rho_B g \Pi, \quad B = \rho g \Pi, \quad F_D = \frac{1}{2} \rho C_d V^2 A \quad (7.7)$$

where ρ_B is the density of the body, ρ is the density of water, Π is the volume of the body, C_d is the drag coefficient, $V = dw/dt$ is the velocity, and A is an effective cross-sectional area normal to the flow. The vertical motion is governed by the equation of motion

$$(\rho_B \Pi + m) \frac{d^2 w}{dt^2} = \frac{1}{2} \rho C_d \left(\frac{dw}{dt} \right)^2 A - (\rho_B - \rho) g \Pi \quad (7.8)$$

where m is the added mass of the fluid. This approach can be used when the cylinder falls vertically with its axis either perpendicular or parallel to the surface of the water.

In the impact mine prediction method (IBPM) developed by Arnone and Bowen [293] and Satkowiak [294,295], the body is assumed to move at a constant attitude (angle ψ between the axis of the cylinder and the horizontal plane) in each medium (air, water, and sediment) but changes as it passes from one fluid to the other. Chu [292] summarizes how to include the effect of ψ on the forces acting on the body, and the motion is then predicted by solving the Kirchhoff–Kelvin equation.

Hurst [296] improved the method by including horizontal and vertical motion of the center of mass and the rotation of the body in that plane. The gravity force is applied at the center of mass (COM), while the buoyancy force is applied at the center of volume (COV). In general, these two points do not coincide and that

results in a change in orientation of the cylinder. This two-dimensional approach was not completely satisfactory for several reasons, one of them being that it does not model the existing currents in the water.

Three-dimensional approaches were developed by Chu et al. [297–311] at the Naval Postgraduate School and another group at MIT [312–314]. Experimental results were presented by researchers at the U.S. Naval Research Laboratory [315–322]. While excellent predictions are obtained under controlled conditions, there are many sources of uncertainties about many of the factors affecting the motion of the cylinder in the air, in the water, and in the sediment. Some studies (e.g., Barbu et al. [322]) adopted a probabilistic approach to analyze the experimental results.

This problem is similar to the hull slamming problem, in the sense that it involves the water entry of a solid, but there are several significant differences: (1) the effective density of the body is larger than the density of water and the body does sink to the bottom and (2) the motion of the body consists mostly of the open cavity phase and the closed cavity phases, whereas, for hull slamming, the cavity-forming phase is of interest.

8 Aerospace Applications

A recent survey [323] provides an overview of the literature concerned with water impacts of aerospace structures. This section will discuss: (1) the landing of seaplanes; (2) the ditching of aircraft; (3) the splashdown of space exploration capsules; and (4) the water landing of solid rocket boosters.

8.1 Landing of Seaplanes. Following the pioneering work of von Karman [6] and Wagner [7], many investigators considered problems related to the water landing of seaplanes, including Baterson [324], Benscoter [325], Milwitzky [326–328], Schnitzer [329–332], Schnitzer and Hathaway [333], Smiley [334,335], and Mayo [336–338]. More recently, Kagawa et al. [339] designed floats for an ultralight seaplane. The mass of the plane is less than 225 kg, and the float is 4 m long and has a mass of less than 14 kg.

8.2 Ditching of Aircraft. The term ditching refers to the emergency water landing of an aircraft not designed for that purpose. It is a planned event in which the crew knowingly makes a controlled emergency landing in water. A celebrated case is that of US Airways Flight 1549, which successfully landed in the Hudson River in New York City, Jan. 15, 2009 and all passengers were evacuated without injuries. However, this procedure is fraught with very significant risks, and the outcome is not always so successful. For example, on Aug. 6, 2005, the ATR-72 aircraft on Tuninter Flight 1153 from Bari International Airport in Bari, Italy to Djerba-Zarzis Airport in Djerba, Tunisia ran out of fuel and ditched into the ocean 23 nautical miles (43 kilometers) north-east of the Palermo International Airport. The aircraft broke into three sections upon impact. Of the four crew members and 35 passengers on board, 16 died. Autopsies indicated that many of the dead succumbed to the impact. Autopsies established that eight passengers, who received injuries during the impact, ultimately drowned.

Scale-model investigations of ditching started in the 1940s and 1950s and led to recommendations for procedures to be followed [340–344] and for the design of airplanes [345–351]. Figure 36 shows the attitude of an airplane prior to ditching. A landing attitude between 5 deg and 12 deg results in a longitudinal and a normal acceleration. Since the impact occurs near the tail section, it will also generate a moment about the center of gravity of the airplane and a negative angular acceleration. The risk of diving upon impact is worse if the landing gear is extended. Generally, there is a risk of damage to the fuselage, since its lower half is subjected to very high pressures, which in turn can lead to sinking in a very short time and limit the possibility of a successful evacuation. Ditching often results in large accelerations in both

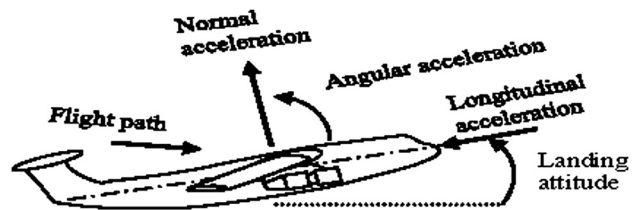


Fig. 36 Ditching of airplanes (Thompson [350])

the longitudinal and vertical directions that can cause injuries to the passengers.

Leigh [352] proposed a momentum approach to estimate aircraft ditching loads that can be considered to be an extension of von Karman's approach. In recent years, numerical methods have been used in the analysis of aircraft ditching. Streckwall et al. [353] used a RANS solver to determine the loads acting on a fuselage during ditching. Many studies are concerned with the crashworthiness of helicopters during impacts against either hard surfaces or water or soft soil. It was found [354] that rigid surface impacts introduce loads directly into beams of the subfloor structure, while impacts on water and soft soil introduce distributed loadings to the skin.

Delsart et al. [355] modeled the water impact of a helicopter using the finite element method. The fluid model was 18 m long, 5 m wide, and 3.5 m deep (2 m of water and 1.5 m of air above it). The model included 715,000 three-dimensional elements for the fluid domain. In Delsart et al. [356], the fluid is modeled using SPH method in a region surrounding the penetrating solid, and the finite element method is used for the remaining of the fluid domain. Examples of a rigid wedge, a rigid cylinder, and a full size helicopter impacting the water are presented. Campbell and Vignjevic [357] analyzed the impact of helicopter subfloor with the surface of water using an SPH model. Hua et al. [358] studied the ditching of a commercial airplane's finite elements with the arbitrary Lagrange–Euler (ALE) technique for the fluid domain. Comparisons of numerical results obtained using the SPH method and experimental results were presented by Anghileri et al. [359–361] and others [362–367]. Several authors have studied the possible ditching of a space shuttle [368–371] and the water impact of the passenger compartment of the Challenger space shuttle [372,373].

For helicopters used in support of offshore oil and gas exploration and production, “ditching” is defined to be a deliberate emergency landing on water [374]. Breaking waves can then cause the helicopter to capsize [375,376]. A review of research related to water impact of helicopters is given in Hughes and Campbell [377]. Design requirements for water impact of helicopters are discussed by Wittlin and Gamon [378].

Experimental and numerical studies of helicopter floors during impact with water were presented in several studies [375–395]. Helicopters designed for sea landings are often equipped with inflated floaters. The numerical simulation of Aksenov et al. [396] dealt with the dynamics of the water landing of the helicopter, while Halbout et al. [397,398] focused on the deformation of the inflated floaters. A new concept using unfolding composite “tensor skins” was developed for increased crashworthiness of helicopter fuselages during water impacts [399,400]. While the term ditching refers to intentional water landings, unintended landings are called crash landings. Fasanella et al. [401,402] studied the crash of a 5-ft-long and 5-ft-diameter fuselage section on a rigid surface and on water through testing and finite element simulation. During a crash, the rigid floor crushes five foam blocks. It was found that a “rigid” impact surface does not absorb any of the kinetic energy, but that, for the water impacts, the water absorbed the major portion of the kinetic energy of the test article. The skin is then subjected to large pressures directed inwards that can cause failure. In addition, Fasanella et al. [403] included test and simulation results for crash impacts on sand. This problem was also

analyzed using two different numerical methods [404], and it was shown that, for the water entry of a rigid hemisphere, the SPH method requires a very fine mesh with over 1×10^6 particles and over 15 h of computational time.

8.3 Splashdown of Apollo Capsules and Similar Structures.

As space exploration started, astronauts traveling in a space capsule reentered the Earth's atmosphere and ended their trip with a water landing. The pressures applied to the vehicle and the accelerations its occupants are subjected to during this type of landing are carefully considered to ensure a successful outcome. Many investigators have studied the effects of accelerations on the human body from the 1950s [405] and 1960s [406–408] to the present [409]. Such impacts affect the nervous, cardiorespiratory, and musculoskeletal systems.

In 1959, tests were conducted [410] to study the water landing of a manned orbital capsule with a conical shape and a spherically shaped bottom. The test article had a 7-ft-diameter base, weighed 1172 lb, and had a 30-ft/s velocity. The objective was to measure the accelerations during the water entry phase. Similar tests were conducted by McGehee et al. [411] and Vaughan [412]. Benson [413] and Jensen and Rosenbaum [414] considered oblique impacts. Figure 37 shows three examples, in which the flight path, defined as the direction of the center of mass of the vehicle, makes a 65 deg angle with the surface of the water. Therefore, the impact velocity will have both a vertical and a horizontal component. In addition to that, the attitude of the vehicle, defined as the orientation between the axis of the vehicle and the vertical, can be either negative, zero, or positive.

Baker et al. [415] discussed scale model development and testing to simulate water entry of Apollo capsules. It should be noted that, in this application, both vertical and horizontal components are in the 30–40-ft/s range and that the axis of the capsule makes an angle of 15 deg to 20 deg from the vertical. Even though that structure is axisymmetric, the determination of the wetted area is not simple and the analysis of the water entry is also challenging. Stubbs [416] and Stubbs and Hathaway [417] presented experimental results for the water impact of Apollo capsules. It was shown that, in certain cases, the capsules roll over. Li and Sugimura [418] presented an analysis of the oblique impact of the Apollo command module on water using a rigid sphere as a mathematical model. Cappelli et al. [114,116,419] considered the deformation of the spherical bottom of the capsule, as discussed in Sec. 6.

More recently, NASA considered water splashdown of similar vehicles numerically and showed that those vehicles can roll over during that phase, which would greatly complicate the evacuation of the crew [420]. Hiraki et al. [421] considered a similarly shaped capsule. Results obtained using von Karman [6] and Wagner [7] approaches and a numerical analysis were compared with experimental results. Wang and Lyle [422] presented a numerical analysis of the water landing of a conceptual space capsule. Brooks and Anderson [423] presented experimental results for a space capsule impacting the free surface of water with both vertical and horizontal velocity components. Lorenz [424] discusses experimental results on the splashdown of a small probe. Modeling is done using the conservation of momentum approach.

8.4 Water Landing of Solid Rocket Motors. The space shuttle solid rocket boosters (SRBs) are the two large reusable rockets used by the space shuttle during the first two minutes of powered flight. After separation, parachutes are deployed and the cylindrical SRBs impact the water surface at 81 ft/s, while their axes remain vertical. Each empty casing has a mass of approximately 91,000 kg.

Buyukozturk et al. [425] performed a finite element analysis of SRBs, assuming a static pressure distribution following a cosine distribution around the circumference. This pressure distribution was assumed to be a first approximation of the loads experienced during water impact. In Hertings' finite element analysis [426], the fluid do-

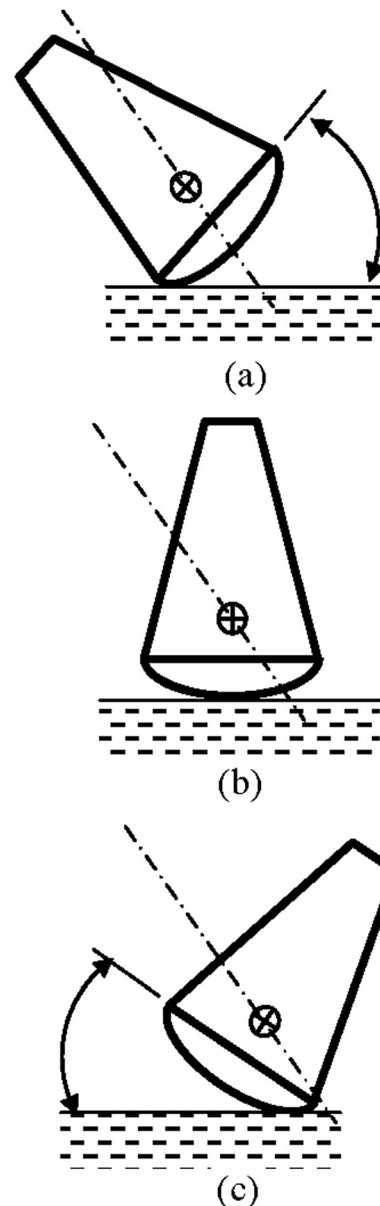


Fig. 37 Attitudes of space capsule before water landing (McGehee et al. [411]). (a) Positive attitude, (b) zero attitude, (c) negative attitude. Dash-dot line: flight path.

main and the shell were considered simultaneously. SRBs impact the ocean at 20–26 m/s (65–85 ft/sec). As they enter the ocean nozzle first, the water moves away from the vehicle, creating a cavity, which then collapses on the vehicle and results in a significant loading event, as shown by Schmidt et al. [427]. Melis [428] presented a finite element analysis of the SRB aft skirt during splashdown. Mitchell and Mahadevan [429] analyzed the forward skirt of SRBs during splashdowns using a combination of a rigid body model and finite element analysis to predict the stresses resulting from the impact.

When a SRB impacts the water, the loads it is subjected to depend of the velocity and the impact angle but also on the weather conditions. High winds and high seas will result in worse case scenarios, but designing for this eventuality leads to high costs. Designing for less severe conditions will reduce the cost of each SRB, but failure will require replacement.

9 Hydrodynamic Ram and Supercavitating Projectiles

Two other types of solid-liquid impacts are hydrodynamic rams and supercavitating projectiles. The first problem often refers to

the ballistic impact of an aircraft fuel tank by projectiles that may or may not explode inside the tank. Other scenarios have also been considered. Supercavitating projectiles or supercavitating vehicles are projectiles equipped with a solid rocket motor that move at high subsonic speeds or supersonic speeds in the water. Cavitation occurs when the water pressure is lowered below its vapor pressure. A super cavity is a single bubble containing the vehicle completely. The cavity reduces the drag on the projectile. The common feature of these two problems is that the motion of the projectile in the water is, after an initial phase, of a cavity-running type as opposed to the cavity-forming type for the hull slamming problem.

9.1 Hydrodynamic Ram. For aircraft, hydrodynamic ram is a phenomenon that has been investigated for a long time, because it is a major damage mechanism in fuel tanks impacted by high-speed projectiles. As it travels through the fluid in the tank, the projectile generates pressure waves. Pressures can exceed 2500 psi in the near vicinity of the projectile, and for high-explosive incendiary projectiles, pressures can exceed 6000 psi after detonation [430]. A review of the early work on that subject was published by Fry [431] in 1976. The entire process can be considered to have three phases: (1) impact and perforation of the front wall of the tank; (2) travel through the fluid; and (3) perforation of the back wall and exit of the projectile.

Characteristics of the running cavity generated by the water impact of a rigid body is discussed in detail by May [110]. Early studies showed that, for the vertical entry of spheres, cavity closure is affected by several factors: (1) surface closure occurs more rapidly for higher entry velocities; (2) the time of surface closure is inversely proportional to the impact velocity [432]; and (3) the time to closure is proportional to the sphere diameter [433]. Then, the dimensionless time to surface closure T_{sc} , defined as

$$T_{sc} = t_{sc} V_o / d \quad (9.1)$$

is nearly constant. In Eq. (9.1), t_{sc} is the time to closure, V_o is the impact velocity, and d is the diameter of the sphere. Duclaux et al. [434] presented experimental results and mathematical models for predicting the pinch distance and time to closure.

The shape of the early water-entry cavity is a parabola [110] and based on four simplifying assumptions (vertical water entry, constant missile speed, cavity pressure equal to ambient pressure, and absence of gravity). The ideal cavity shape is given by

$$\frac{x}{C_D^{1/2} d} = 1.716 \left(\frac{y}{C_D^{1/2} d} \right)^2 \quad (9.2)$$

McMillen [111] studied the shock wave produced by the impact of 3.175–6.35-mm spheres entering water with velocities between 610 and 1463 m/s. Results from this work are important in understanding the travel of the projectile through the fluid. Figure 38 is a shadow graph taken during one of his experiments and shows a spherical wave front and the cavity generated by the projectile. It was shown that the pressure on the wave front varies with the position defined by the angle θ . It is maximum when $\theta = 90$ deg. For a projectile moving in the 90 deg direction with a velocity V , the normal velocity in the θ direction (Fig. 39) is $V \sin \theta$. In linear acoustics, the pressure is equal to the velocity times the mechanical impedance. Therefore, one could expect the pressure to vary according to $\sin \theta$. McMillen [111] showed that, when $\theta < 75$ deg, a better fit of the experimental data is obtained with

$$p = p_{90} \sin(\theta - 7 \text{ deg}) \quad (9.3)$$

The pressure was shown to increase linearly with the projected area of the sphere and to be directly proportional to $V^{2.17}/R$. In the experiments of Disimile et al. [435], spherical projectiles made

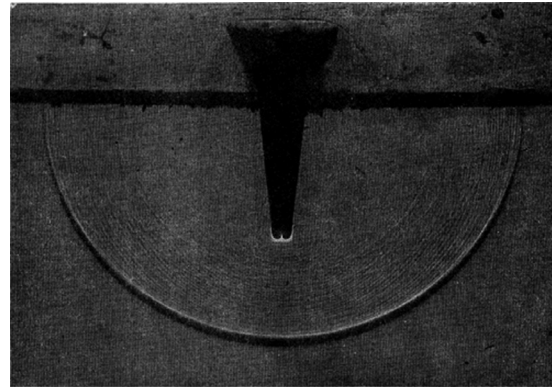


Fig. 38 Hemispherical wave in water after a $\frac{1}{8}$ -in-diameter ball impacted the surface and penetrated 7.8 cm. The velocity at that instant is 3520 ft/s (1072.9 m/s), which is 35% less than the impact velocity (McMillen [111], reproduced with permission from the American Physical Society).

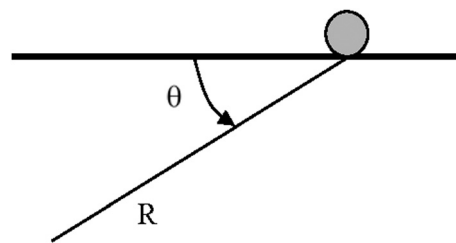


Fig. 39 Polar coordinates used in Eqs. (9.3) and (9.4) for the impact of a sphere on the water surface

out of aluminum, steel, and tungsten carbide were used. These projectiles have a 12.7-mm diameter and their masses were 2.79, 7.85, and 14.29 g, respectively. The impact velocities were around 350 m/s. It was found that the pressure at the wave front varied with the angle as

$$p = p_{90} \sin(\theta + 7 \text{ deg}) \quad (9.4)$$

Disimile et al. [435] also showed that, in their experiments, the pressure decayed with distance a little bit faster than $1/R$. Shock wave velocities depend on the strength of the shock, so that the wave front is expected to have an elliptical shape with its major axis in the direction of motion of the projectile. McMillen [111] showed that the wave front can be considered to be spherical for velocities below 1220 m/s, and later he showed that, behind the shock wave, there are secondary waves produced by secondary vibrations of the sphere [436]. The experiments of Shi and Kume [437,438] with small impact velocities (28 m/s) are in general agreement with the findings of McMillen and other researchers with regards to the shock wave propagating through the fluid.

Lecysyn et al. [439–441] considered the effects of ballistic impacts on cylindrical steel vessels with 360-mm diameters, heights of 622 mm, thicknesses of 0.7 mm, and a volume of 60 L. In industry, these vessels could contain many kinds of fluids with a wide range of density and viscosity, such as water, oil, fuel, solvents, and bitumen, for example. In these studies, the barrels were not completely filled; the gap between the liquid surface and the top of the barrel was 40 mm. The projectiles were solid spheres with a 13-mm diameter, made out of a tungsten alloy called denal. The mass of the projectiles ranged from 20.1 g to 20.2 g, and the impact velocities were in the 1000–1500 m/s range.

The cavity-running phase can be modeled using Newton's law,

$$m \frac{dv}{dt} = -\frac{1}{2} \rho A C_d v^2 \quad (9.5)$$

where m is the mass of the projectile, ρ is the density of the fluid, A is the projected area of the projectile, C_d is the drag coefficient, and v is the velocity of the projectile. Richardson [112] showed that $C_d = 0.4$, with a contact angle of 115 deg. This value of C_d is used by Varas et al. [442,443], who studied cases in which a 12.5-mm steel sphere with a mass of 8 g impacts a tank with velocities of 600 or 900 m/s. Assuming that the drag coefficient remains constant, this equation can be integrated by first rewriting it as

$$-\frac{dv}{v^2} = \beta dt \quad (9.6)$$

with $\beta = \frac{1}{2} \rho A C_d / m$. For a spherical projectile, $\beta = \frac{3}{4} (\rho / \rho_p d_p) C_d$, where ρ_p is the density of the sphere and d_p its diameter. Then, we find that the velocity decays with time according to

$$v = V / (1 + \beta V t) \quad (9.7)$$

Similarly, since $(dv/dt) = (dv/dx)(dx/dt) = v(dv/dx)$, Eq. (9.5) can be written as

$$\frac{dv}{v} = -\beta dx \quad (9.8)$$

and, after integration, the velocity of the projectile is found to decay exponentially with position

$$v = V e^{-\beta x} \quad (9.9)$$

as mentioned by Refs. [441] and [444]. The experiments of Shi and Takami [445,446] and Lee [447,448] showed that the velocity decayed exponentially with the penetration depth, as predicted by Eq. (2.32). Integrating Eq. (9.7) gives the position of the projectile as a function of time

$$x = \frac{1}{\beta} \ln(1 + \beta V t) \quad (9.10)$$

The experimental results of Petitpas et al. [449] for a projectile penetrating a water-filled tank at 1243 m/s are in excellent agreement with the predictions from Eq. (9.10) with a drag coefficient $C_d = 0.41$. Figure 40 shows that the shock wave is moving faster than the projectile, but the wavefront is not spherical, as in McMillen's experiments (Fig. 41), since the projectile does not enter the water directly but only after perforating the wall of the tank. The motion of that wall affects the shape of the wave front.

Rand et al. [450] used this model to study the longitudinal motion of the projectile and introduce a model for the rotational

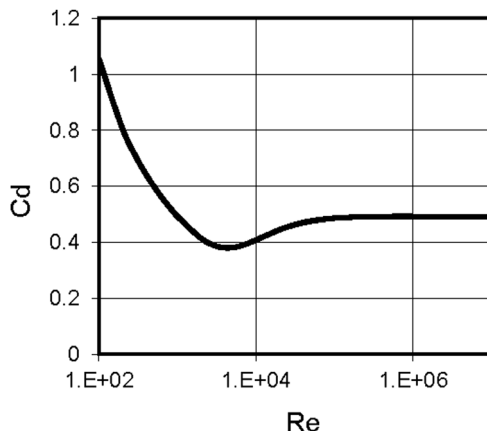


Fig. 40 Drag coefficient C_d as a function of Reynolds number Re (Eq. (9.11))

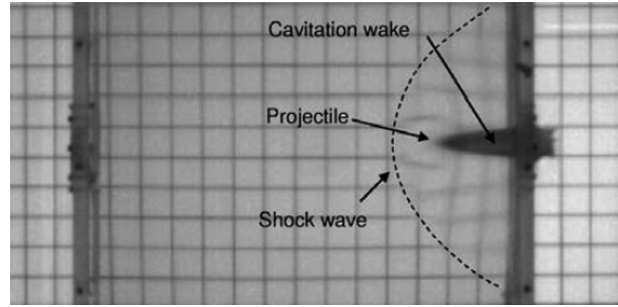


Fig. 41 Shock wave generated by the impact of a projectile on a water-filled tank with an initial velocity of 1243 m/s (Petitpas et al. [449], reproduced with permission from Elsevier)

motion for studying the slapping of the projectile's tail against the cavity. The projectile is assumed to be pinned at the nose, and the impact against the cavity wall is assumed to be elastic with a coefficient of restitution equal to one.

Lecysyn et al. [439,441] also considered the case when the drag coefficient varied according to

$$C_D = \frac{24}{Re} + \frac{3.73}{\sqrt{Re}} - \frac{4.83 \times 10^{-3} \sqrt{Re}}{1 + 3 \times 10^{-6} Re^{3/2}} + 0.49 \quad (9.11)$$

and integrated Eq. (9.5) numerically using Euler's method. Figure 41 shows that, according to this formula, the drag coefficient remains constant for Reynold's numbers above 100,000.

Additional numerical studies of high velocity impacts of fluid-filled containers were presented in Refs. [451–458]. Varas et al. [459] studied fluid-filled tubes made of carbon-reinforced composite materials subjected to high velocity impact. Disimile et al. [460] analyzed the effect of a system of triangular bars inside a large water tank on the shock wave produced by a hydrodynamic ram.

9.2 Supercavitating Projectiles. Another water impact problem is that of underwater missiles moving underwater at high velocity. Typically, these missiles consist of a main body containing a solid combustible fuel and a nozzle through which combustion gases escape at the rear. Ahead of the main body is a smaller body, called a cavitator, which pierces the water and creates a cavity ahead of the missile. Often the cavitator is a small plate oriented at an angle from the axis of the missile in order to generate a vertical force to balance the weight of the missile. The cavity is initially filled with a liquid-vapor mixture. The principle is to create a cavity sufficiently large, so that the main body is not in contact with the water. Fins are usually added for stability. Examples are given in Refs. [449] and [461].

Ahn [462] studied small-scale projectiles, such as rapid airborne mine clearance systems (RAMICSS), a supercavitating projectile that travels in air as well as water and is fired from a gun carried by a helicopter, targeting near-surface mines. Another example of supercavitating projectiles is the adaptable high-speed undersea munitions (AHSUMs), which are fired from a submerged gun carried by ships and submarines, travel underwater at speeds around 1500 m/s, and target incoming torpedoes. That study shows several possible scenarios. At low velocities, one cavity is formed in the front, but it is not large enough for the entire missile, and a second cavity forms at the rear due to the exhaust gases. At higher velocities, a fully developed supercavity is formed, and the projectile can drag its tail on the bottom of the cavity (planing), be tail-slapping inside the cavity, or move with aerodynamic interaction inside the cavity.

The cavity for a supercavitating projectile is a streamline, along which, according to the Bernoulli equation, the pressure p can be calculated using [463]

$$p = p_o + \frac{1}{2}\rho(U^2 - u_s^2) \quad (9.12)$$

where p_c is the cavity pressure, U is the velocity of the projectile, and u_s is the velocity of the water along that streamline. On the surface of the cavity, the pressure is equal to p_c , the vapor pressure inside the cavity, and, in that equation, $p_o = \rho gh$ is the static pressure at the depth h below the surface of the water. Solving for u_s , we find that

$$u_s = U\sqrt{1 + \sigma} \quad (9.13)$$

where σ is the cavitation number defined by

$$\sigma = (p_o - p_c) / \left(\frac{1}{2}\rho U^2 \right) \quad (9.14)$$

The shape of the cavity has been studied by many authors, and the early work in this area has been discussed in detail by May [110]. Nowadays, several models are used: the Munzer–Reichardt model described by Ahn [462], Logvinovich's model [464–468], and that proposed by Savchenko [469]. The Munzer–Reichardt's model predicts an axisymmetric cavity shape described by

$$r_c = \frac{d_{\max}}{2} [4\zeta(1 - \zeta)]^{1/2.4} \quad (9.15)$$

where r_c is the radius of the cavity, $d_{\max}$ is its maximum diameter, $L_{\max}$ its length, and $\zeta = x/L_{\max}$. The distance x is measured from the projectile's nose. The dimensions of the cavity are given by

$$d_{\max} = d_c \sqrt{\frac{C_D(\sigma, 0)}{\sigma}}, \quad L_{\max} = d_c \sqrt{\frac{C_D(\sigma, 0)}{\sigma^2} \ln\left(\frac{1}{\sigma}\right)} \quad (9.16)$$

For a disk cavitator at an angle of attack α_c , the drag coefficient is given by

$$C_d(\sigma, \alpha_c) = C_{do}(1 + \sigma) \cos^2 \alpha_c \quad (9.17)$$

where $C_{do} = 0.815$. Kulkarni and Pratap [470] assumed that the cavity has an ellipsoidal shape, with the length of the axes given by Eq. (9.16).

High-speed imaging of supercavitating underwater projectiles [471] showed excellent agreement between the experimentally determined cavity shape and the predictions from the Munzer–Reichardt theory. It also provided experimental evidence of the shape of the wave front for a projectile traveling at supersonic speed. Figure 42 illustrates the formation of a shock front for a projectile moving at supersonic speeds. The projectile moves with a velocity V and, following Huyghens principle, at each instant, a spherical wave expands with a velocity c . The resulting wave front is a conical surface with the Mach angle

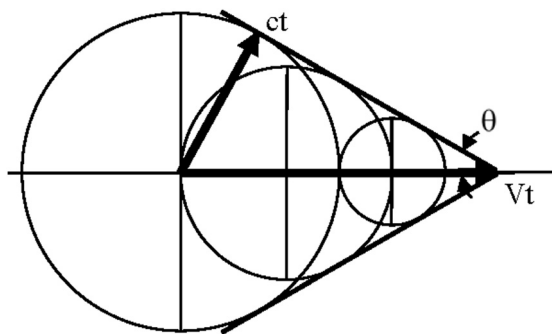


Fig. 42 Formation of a wave front for a supersonic projectile

$$\theta = \sin^{-1} \frac{a}{V} = \sin^{-1} \frac{1}{M} \quad (9.18)$$

where M is the Mach number. Equation (9.18) is used to estimate the Mach number from photographs.

Supercavitating vehicles undergo high longitudinal forces as a result of their high underwater velocity. The drag force and the thrust that compress the body in the axial direction may induce buckling. Ruzzene [472,473], Ahn and Ruzzene [474], and Choi and Ruzzene [475] studied the buckling and dynamic stability of shells under axial loads for this particular type of application. The optimization of a supercavitating torpedo is discussed by Alyanak et al. [476].

10 Conclusion

Hull slamming and other water impact problems have been studied extensively over more than a century. This article provides a comprehensive overview of the literature in this area with 477 references from 1882 to 2012. It should be noted that 51% of these references appeared between 2000 and 2012, reflecting increased interest in the subject, major developments in the simulation capabilities, and improvements in experimental techniques for detailed observations.

In hull slamming, the initial velocity of the body is rather small compared to that of ballistic projectiles and supercavitating projectiles. The effective density of the hull is less than that of water as opposed to mines that are expected to fall to the bottom of the ocean and bury themselves in the sediment. Hull slamming is similar to the water landing of seaplanes or the ditching of aircraft in terms of the velocity and the effective density. This is why some of the early work by von Karman [6] and Wagner [7] is applicable to hull slamming and is still used today. The discussion of these other types of problems in Secs. 8 and 9 puts the discussion of hull slamming in a larger context and provides more insight into the fluid-structure interaction problem.

Approximate solutions are available for some simple cases in order to predict velocities, accelerations, and pressure distributions. A general conclusion from this review is that numerical methods are useful when complex geometries and various complicating factors have to be considered. However, it is often difficult to get accurate solutions, and specialized techniques are often required.

A comprehensive list of references is provided, and this report outlines both the physical phenomena and the modeling issues associated with hull slamming and water impact of projectiles in general. Hopefully, it will be useful to all readers from the beginner to the more advanced.

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